The Sharpe Stability Ratio in Tactical and Static Asset Allocation Backtests: Evidence from 173 Published Strategy Variants

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Abstract

A Sharpe ratio says how much risk-adjusted return a backtest earned, not how steadily it earned it. The Sharpe Stability Ratio (SSR) of Bajo Traver and Rodriguez Dominguez (2026) measures the steadiness: the mean of a rolling 36-month Sharpe ratio divided by an autocorrelation-robust estimate of its long-run standard deviation. We compute it for 173 tactical and static asset allocation variants published on one backtesting platform, with 18 to 127 years of monthly history ending in August 2026. Our message is that the SSR adds information to the Sharpe ratio but describes a track record over a given period, not a lasting property of a strategy. It ranks the catalog differently from the Sharpe ratio (rank correlation 0.64) and from the Deflated Sharpe Ratio (0.59): of 146 variants with a Deflated Sharpe Ratio of at least 0.90, 53 have an SSR of at least 0.5 and 93 do not. Its rankings barely move with the window or bandwidth but change from decade to decade (rank correlations of 0.08, −0.38 and 0.11 between adjacent subperiods). Much of its level is the Sharpe ratio itself: the same returns in random order give about 0.4 times the Sharpe ratio, more for shorter histories. The ratio of the actual SSR to that benchmark is unrelated to the Sharpe ratio (rank correlation −0.01) and exceeds one for 127 variants, but none survives a false discovery rate adjustment. Measured above cash, SSR levels fall by about a quarter and the ranking keeps a rank correlation of 0.71 with the zero-reference one: the findings hold in substance, while the leading names change.

1. Introduction

A backtest's Sharpe ratio says how much risk-adjusted return a strategy earned over the sample. It does not say how that return was spread over time: two strategies with similar Sharpe ratios can have earned them steadily, or in a handful of episodes that happened to fall inside the sample. The Probabilistic Sharpe Ratio (Bailey and López de Prado, 2012) asks whether the full-sample Sharpe ratio is distinguishable from a benchmark, and the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) whether it beats the best of many trials. Neither describes how the estimate evolved over time.

The Sharpe Stability Ratio (SSR) of Bajo Traver and Rodriguez Dominguez (2026) fills that gap. It follows a strategy's Sharpe ratio through time, over a rolling 36-month window, and compares the average of that rolling series with how much the series wanders. A strategy whose rolling Sharpe ratio stays close to a high average scores high; one whose rolling Sharpe ratio is high on average but swings in long waves scores low. Two strategies from our sample show the difference. Hybrid Asset Allocation has a full-history Sharpe ratio of 1.50 and an SSR of 0.48; the Momentum-Correlation Triplet has a lower Sharpe ratio, 1.21, and an SSR of 0.94, the highest in the catalog (Section 3).

We compute the SSR for 173 tactical and static asset allocation strategy variants published on BestFolio, a tactical asset allocation (TAA) research platform. Most are reconstructions of published models (Faber, 2007; Antonacci, 2014; Keller and Keuning, 2016, 2017, 2018, 2022, 2023, among others) and the static allocations investors compare them with. Their histories are simulated rather than lived, they are long (499 months at the median, up to 1,519), and every variant is scored on the same month-end series with the same execution assumptions. This lets us examine, on one consistent catalog, how the SSR relates to the Sharpe and Deflated Sharpe ratios, how much it depends on the evaluation period, how much of it the Sharpe ratio alone would produce, and how it relates to reported turnover.

The paper has one message:

> **The SSR adds information that the Sharpe and Deflated Sharpe ratios do not carry, but it describes a track record over a given period, not a lasting property of a strategy.**

Four findings support it.

1. **It ranks strategies differently.** Its rank correlation with the Sharpe ratio is 0.64 and with the Deflated Sharpe Ratio 0.59. Of the 146 variants whose Deflated Sharpe Ratio is at least 0.90, 53 have an SSR of at least 0.5 and 93 fall below it (Section 5.1).
2. **Much of its level is the Sharpe ratio and the length of the history.** The same monthly returns in random order give an SSR of about 0.39 times the Sharpe ratio, and more for shorter histories. The ratio of the actual SSR to this random-order benchmark, the permutation multiple, is unrelated to the Sharpe ratio (rank correlation −0.01) and exceeds one for 127 of the 173 variants, but no single variant's multiple survives a false discovery rate adjustment (Section 5.2).
3. **Its ranking changes from decade to decade.** Rankings barely move with the window or the bandwidth (rank correlations of 0.92 to 0.99 with the baseline), but between adjacent calendar subperiods the rank correlation is 0.08, −0.38 and 0.11, and the S&P 500's own SSR goes from 0.15 in the 2000s to 1.63 in the 2010s (Section 5.3).
4. **High scores go with monthly tactical rotations and with higher turnover, descriptively.** Tactical variants have a higher median SSR than static ones (0.46 against 0.39), and turnover is positively associated with the SSR (rank correlation 0.26), but these comparisons are confounded and identify no cause (Section 5.4).

The estimates are uncertain: a block bootstrap places the 95 percent interval of all but one variant above zero and no variant's 5th percentile above 0.5 (Section 5.5). The findings do not depend on the cash reference: measured above US dollar cash, as the platform now publishes its Sharpe ratio, the levels fall and the leading names change, but the four findings hold in substance, the permutation result more weakly (Section 5.6). Our reading, which follows the statistic's authors, is that the SSR describes a track record and is not a forecast.

The paper extends the SSR from hedge fund indices to a large catalog of simulated asset allocation backtests, shows that its level depends on the length of the history, and adds a permutation benchmark, suggested by Randy Harris of Dual Momentum Systems, that

separates the part of the SSR that the Sharpe ratio explains from the part due to the order of returns.

How to read this paper. Section 2 defines every term. Section 3 explains the SSR with one worked example and states the formula. Section 4 describes the data in brief. Section 5 gives the evidence for the four findings and repeats the headline results above cash (Section 5.6), Section 6 the limitations, Section 7 the relation to the literature, and Section 8 concludes. The appendices hold the details: the estimator, bandwidth and bootstrap (Appendix A), the data, sample and replication (B), the Deflated Sharpe Ratio and its count of trials (C), every results table (D), robustness checks, including the results above cash in full (E), the extended literature review (F) and the version history (G).

2. Key terms

Term	Meaning in this paper
Backtest	A simulated history of a rules-based strategy, produced by the platform's engine on historical prices.
Variant	One parameter setting of a strategy, with its own backtest. The sample has 173.
Family	All variants of one strategy. Variants of a family share rules and often assets, so their returns are correlated.
Tactical, static	A tactical variant changes its allocation according to momentum, trend, breadth, volatility or macroeconomic rules; a static variant holds fixed weights and rebalances on a schedule (for example 60/40).
Monthly return	The simple return from one completed month-end to the next. Returns are used raw, with a *cash reference of zero*: no cash rate is subtracted, as the statistic's authors recommend.
Return above cash	The monthly return less the return of US dollar cash (a Treasury bill fund chain) between the same two month-ends. It is the convention of the platform's published Sharpe ratio and Stability value, and Section 5.6 repeats the headline results with it.
Sharpe ratio	Mean monthly return divided by its standard deviation, annualised by multiplying by $\sqrt{12}$. *Full-history* means over the variant's whole backtest.
Rolling Sharpe ratio	The Sharpe ratio over the last 36 months, recomputed every month: a time series with one value per month, Z_t in (1).
Window	The length of the rolling window, $\tau = 36$ months.
Long-run standard deviation	A standard deviation that also counts how long the series remembers its past (its autocorrelation). Estimated with the *HAC* (heteroskedasticity and autocorrelation consistent) formula of Newey and West (1987), equation (5).
Bandwidth	The number of lags of memory, L, that the long-run standard deviation counts. Here $L = 35$, the full extent of the overlap between 36-month windows.

Term	Meaning in this paper
SSR	Sharpe Stability Ratio: the mean of the rolling Sharpe ratio divided by its long-run standard deviation, equation (6). Higher means a steadier risk-adjusted return relative to its level.
Probabilistic Sharpe Ratio (PSR)	The probability, under its model, that the true Sharpe ratio exceeds a benchmark (here zero), given the sample length and the skewness and kurtosis of returns.
Deflated Sharpe Ratio (DSR)	The PSR with the benchmark raised to the Sharpe ratio expected from the best of K unskilled trials. The platform flags a strategy as fragile below 0.90.
Trials, K	The number of strategies tried, used by the DSR. Here $K = 219$, the active variants of the dated catalog that defines the sample (Section 4, Appendix C).
Rank correlation	Spearman's correlation between the rankings of two statistics across the variants: 1 means the same order, 0 no relation, −1 the reverse order.
Subperiod	A calendar period over which the SSR is recomputed: before 2000, 2000 to 2009, 2010 to 2019, and January 2020 to August 2026.
Random-order benchmark	The mean SSR of a series' own monthly returns put in random order 10,000 times. Reordering keeps the mean, the standard deviation and the Sharpe ratio, and destroys any order in time.
Permutation multiple	The actual SSR divided by its random-order benchmark, equation (8). Above one: steadier than chance ordering.
Permutation p-value	The share of random orderings whose SSR is at least the actual one, equation (9).
False discovery rate	The expected share of false findings among the results declared significant; the Benjamini and Hochberg (1995) adjustment controls it at 5 percent when many variants are tested at once.
Block bootstrap	Resampling a series in contiguous blocks of 36 months, 2,000 times, to show how much the SSR could vary. The *95 percent percentile interval* runs from the 2.5th to the 97.5th percentile of the replicates.
Turnover	The share of the portfolio traded per year, as reported by the platform.

3. The SSR in one example

3.1 The idea

Take a strategy's monthly returns and compute its Sharpe ratio over the 36 months ending in a given month; move the window forward by one month and repeat. The result is the rolling Sharpe ratio, a time series with one value per month. Figure 1, panel A, shows it for Hybrid Asset Allocation (HAA) and the Momentum-Correlation Triplet (the Triplet).

Two features of that series matter. The first is its level. On average HAA's rolling Sharpe ratio is higher than the Triplet's: 1.58 against 1.18, annualised (the dashed lines). The second is how

far, and for how long, the series strays from its level. HAA's rolling Sharpe ratio was far above its average in its first decade and below it for long stretches afterwards; the Triplet's stays closer to its average and returns to it faster. The SSR divides the first feature by a measure of the second.

That measure needs care. Neighbouring values of the rolling Sharpe ratio share 35 of their 36 months, so the series is smooth by construction, and its ordinary standard deviation understates how far its average could be from the truth. Panel B shows this memory as the autocorrelation of the rolling series at each lag k. The dotted line is what the overlap of the windows alone would produce, $1 - k/36$ (Bajo Traver and Rodriguez Dominguez, 2026, Proposition 3.1). HAA's rolling series remembers more than the overlap explains, beyond 36 months; the Triplet's forgets faster. A long-run standard deviation counts this memory, and the ordinary one does not.

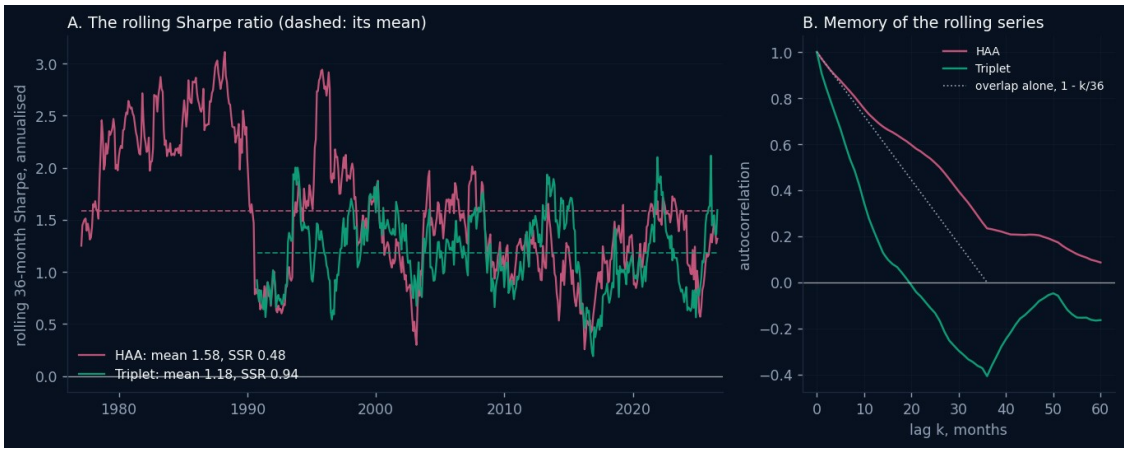


Figure 1. The worked example. A: rolling 36-month annualised Sharpe ratio of Hybrid Asset Allocation (from February 1977) and the Momentum-Correlation Triplet (from October 1990) through August 2026, with each series' mean (dashed). B: autocorrelation of each rolling series by lag, against the autocorrelation that the overlap of 36-month windows alone produces.

3.2 The formula

Let $r_1, \ldots, r_T$ be a strategy's monthly returns. The rolling Sharpe ratio at month t is

$$Z_t = \frac{\bar{r}_t}{s_t}, t = \tau, \ldots, T \tag{1}$$

with the mean and the sample standard deviation of the τ returns ending in month t,

$$\bar{r}_t = \frac{1}{\tau}\sum_{j=0}^{\tau-1} r_{t-j}, s_t^2 = \frac{1}{\tau-1}\sum_{j=0}^{\tau-1}\left(r_{t-j} - \bar{r}_t\right)^2 \tag{2}$$

The rolling series has $N = T - \tau + 1$ values. Its mean, its autocovariance at lag k and its long-run variance are

$$\bar{Z} = \frac{1}{N} \sum_{t=\tau}^{T} Z_t \tag{3}$$

$$\hat{\gamma}_k = \frac{1}{N} \sum_{t=\tau+k}^{T} \left(Z_t - \bar{Z}\right)\left(Z_{t-k} - \bar{Z}\right) \tag{4}$$

$$\hat{\sigma}_L^2 = \hat{\gamma}_0 + 2 \sum_{k=1}^{L} \left(1 - \frac{k}{L+1}\right) \hat{\gamma}_k \tag{5}$$

and the Sharpe Stability Ratio is

$$\mathrm{SSR} = \frac{\bar{Z}}{\hat{\sigma}_L} \tag{6}$$

where

- r_t is the simple return in month t, from completed month-end values, with a cash reference of zero;
- τ is the window, 36 months, and T the number of monthly returns;
- $\bar{r}_t$ and s_t are the mean and the sample standard deviation of the 36 returns ending in month t;
- Z_t is the rolling Sharpe ratio per month (multiply by $\sqrt{12}$ to annualise; annualisation cancels in (6));
- N is the number of rolling estimates and $\bar{Z}$ their mean;
- $\hat{\gamma}_k$ is the autocovariance of the rolling series at lag k, and $\hat{\gamma}_0$ its ordinary variance;
- L is the bandwidth, 35, and $1 - k/(L+1)$ are the Bartlett weights of Newey and West (1987);
- $\hat{\sigma}_L$ is the long-run (HAC) standard deviation of the rolling series.

Equation (5) is the ordinary variance plus a weighted sum of the autocovariances, which is how the memory of Figure 1, panel B, enters the statistic. The SSR uses a benchmark Sharpe ratio of zero, as its authors recommend, and is reported only for histories of at least 120 months. Appendix A gives the estimator in full: why the bandwidth is fixed at 35, how this fixed-bandwidth version relates to the structural ratio of the source paper (it sits roughly 22 percent above it under the mechanical-overlap approximation), the reporting gate, and the bootstrap.

3.3 Worked example

Table 1 puts the two strategies of Figure 1 through equations (3) to (6).

Table 1. Worked example: the SSR of HAA and the Triplet, full history through August 2026. Values per month unless stated.

	HAA	Triplet
First month of the rolling series	February 1977	October 1990
Rolling estimates, N	595	431
Full-history Sharpe ratio, annualised	1.50	1.21
Mean rolling Sharpe ratio, annualised	1.58	1.18
Mean rolling Sharpe ratio per month, $\overline{Z}$ (annualised value divided by $\sqrt{12}$)	0.457	0.342
Ordinary standard deviation of the rolling series	0.184	0.104
Long-run standard deviation, $\hat{\sigma}_{35}$	0.948	0.365
SSR $= \overline{Z}/\hat{\sigma}_{35}$	0.48	0.94

HAA starts with the higher average, 0.457 per month against 0.342. But its long-run standard deviation, 0.948, is five times its ordinary standard deviation, because its rolling Sharpe ratio drifts in waves longer than the window; the Triplet's long-run standard deviation is 0.365. Dividing, HAA's SSR is 0.457/0.948 = 0.48 and the Triplet's 0.342/0.365 = 0.94. A low SSR does not mean losses: all 595 of HAA's rolling estimates are positive, with a minimum annualised value of 0.26. It means that HAA's risk-adjusted return was concentrated in some decades more than in others. On the window since 2000, HAA's SSR is 0.87 (Table D.4), a first sign of the period dependence of Section 5.3.

3.4 Reading an SSR value

A higher SSR means a steadier rolling Sharpe ratio relative to its level. The SSR is not a probability and has no calibrated threshold; in this catalog it runs from 0.09 to 0.94 with a median of 0.45, and we use 0.5 as a descriptive cutoff. Its level depends on four choices that must be stated with any value: the window, the bandwidth (with 5 lags instead of 35 the median SSR is 0.85 instead of 0.45), the length of the history (Section 5.2) and the cash reference (above cash the median is 0.32, Section 5.6). Rankings are much less sensitive to these choices than levels are.

4. Data in brief

The sample is defined on the BestFolio catalog as it stood on 5 September 2026: 219 active variants, of which 177 were published on the public leaderboard and 42 were research variants, tested but not published. The series are those of the same variants as re-run by the platform on 17 October 2026, after an audit of its proxy chains: indicators now warm up on price history before a strategy's earliest permitted start, so most backtests begin at that start (typically January 1985 rather than early 1986), and several proxy histories were corrected. Positions drift between signals and fund histories are total return, as since the 28 September 2026 restatement. The primary sample is the published set less three variants whose macroeconomic input was reconstructed with revised rather than point-in-time data, and less one variant with fewer than 120 monthly returns: 173 variants from 92 strategy families, 142 tactical and 31 static. Returns run through 31 August 2026 (completed months only); histories start between 1900 and 2008, with a median of 499 months, a shortest of 220 and a longest of 1,519 (an all-

equity US allocation that the platform now extends back to February 1900). The platform's engine simulates every backtest daily, with one-way transaction costs of 10 basis points and execution on the session after a signal, and substitutes documented total-return proxies where an instrument's history is too short. The benchmarks are the S&P 500 (the Vanguard 500 Index Fund, from 1976), US bonds, a 60/40 blend of the two, gold, long Treasuries and the Nasdaq 100. Appendix B gives the full description, the proxy chains and the replication archive.

The Deflated Sharpe Ratio is computed with the platform's own implementation on the paper's trial set, $K = 219$: the active variants of the dated 5 September 2026 catalog that defines the sample. The platform's live scores count every variant with a stored backtest, active or retired, except eight single-fund holdings that apply no rule, the rule in force since 18 October 2026, when the platform also began scoring above cash: applied to the database of 17 October 2026 it gives 278 trials (Appendix C.1). We keep $K = 219$ so that every number in the paper refers to the same dated cut, and say so wherever the score appears; with the live count the no-skill benchmark is 0.60 rather than 0.58, the scores differ by 0.001 at the median and by at most 0.061, and three variants fall below 0.90 (Appendix C).

5. Evidence

5.1 The SSR ranks strategies differently from the Sharpe and Deflated Sharpe ratios

Across the 173 variants the SSR runs from 0.09 to 0.94, with a median of 0.45; 55 variants reach 0.5. The minimum is an estimator artefact of one near-cash episode (Appendix D.4). The S&P 500 scores 0.32 and the 60/40 blend 0.34, both in the bottom quartile of the catalog (Table D.1, Figure D.1).

Table 2 and Figure 2 compare the SSR with the companion statistics. The rank correlation with the Sharpe ratio is 0.64: strategies with high Sharpe ratios tend to have high SSRs, but the order is far from identical, as HAA and the Triplet show. The rank correlation with the Deflated Sharpe Ratio is 0.59. The SSR is only weakly related to the level of return (0.15 with the compound annual growth rate) and moderately to the depth of the worst drawdown (0.54; shallower drawdowns go with higher SSRs).

Table 2. Spearman rank correlation of the full-history SSR with companion statistics, 173 published variants.

Companion statistic	Rank correlation with SSR
Sharpe ratio	0.64
Deflated Sharpe Ratio	0.59
Probabilistic Sharpe Ratio at zero	0.46
Maximum drawdown (less negative is higher)	0.54
Annual turnover	0.26
Compound annual growth rate	0.15

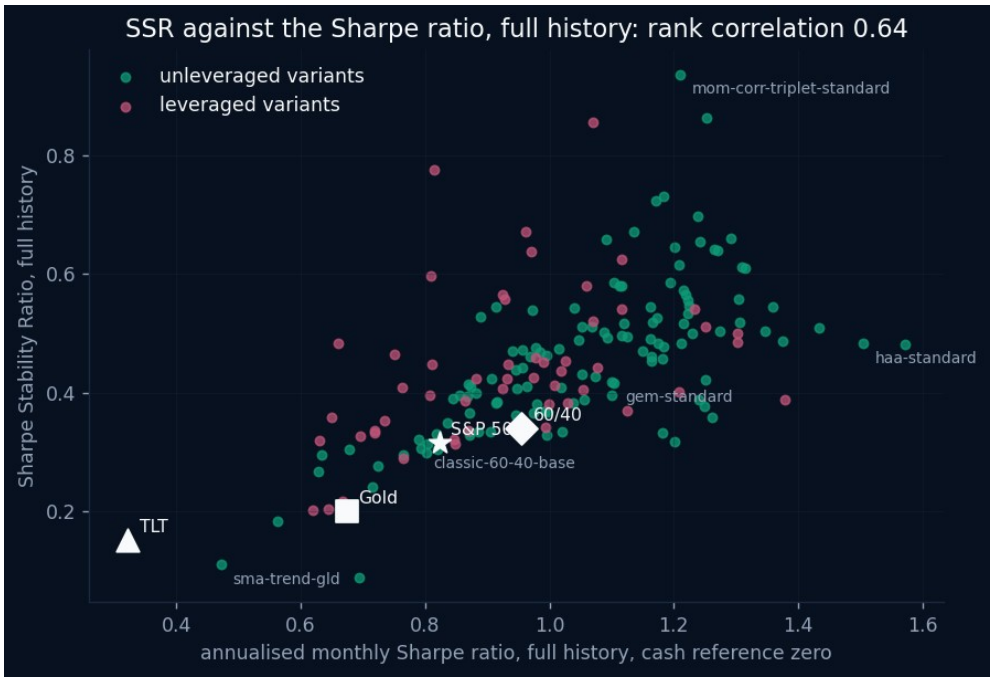


Figure 2. Full-history SSR against the annualised monthly Sharpe ratio (zero cash reference), 173 published variants, benchmarks marked. Rank correlation 0.64.

Table 3 shows what this means at the thresholds the platform uses. Of the 173 variants, 146 have a Deflated Sharpe Ratio of at least 0.90, the platform's cutoff for flagging a strategy as fragile. Of those 146, 53 have an SSR of at least 0.5 and 93 fall below it. The SSR also separates strategies that the Deflated Sharpe Ratio, as commonly displayed, cannot: 89 variants have a Deflated Sharpe Ratio that rounds to 1.00, and at two decimals the SSR takes 52 distinct values against 28. Neither classification establishes prospective credibility; they answer different questions.

Table 3. Joint classification of the 173 published variants by the Deflated Sharpe Ratio ($K = 219$, threshold 0.90) and the full-history SSR (threshold 0.5).

	SSR at or above 0.5	SSR below 0.5
DSR at or above 0.90	53	93
DSR below 0.90	2	25

With the platform's live count of trials (278) the four cells would hold 53, 90, 2 and 28 (Appendix C).

5.2 Much of the level is the Sharpe ratio and the length of the history

How much of a variant's SSR would arise from rolling estimation alone, with no order in its returns? A direct benchmark, suggested by Randy Harris of Dual Momentum Systems, keeps each series' own monthly returns and discards their order. Reordering leaves the mean, the standard deviation and the full-sample Sharpe ratio unchanged, and removes serial dependence, volatility clustering and regime structure. For each series we draw $B = 10,000$ random orderings and compute

$$\overline{\mathrm{SSR}}_{\mathrm{rand}} = \frac{1}{B}\sum_{j=1}^{B} \mathrm{SSR}^{(j)} \tag{7}$$

$$M = \frac{\mathrm{SSR}}{\overline{\mathrm{SSR}}_{\mathrm{rand}}} \tag{8}$$

$$p = \frac{b+1}{B+1} \tag{9}$$

where $\mathrm{SSR}^{(j)}$ is the SSR, equations (1) to (6), of the j-th random ordering, $\overline{\mathrm{SSR}}_{\mathrm{rand}}$ is the random-order benchmark, M is the permutation multiple, and b is the number of orderings whose SSR is at least the actual one, so that p is the one-sided permutation p-value (Phipson and Smyth, 2010).

The benchmark scales with the Sharpe ratio. Across the 173 variants, a regression of the random-order benchmark on the Sharpe ratio through the origin has a slope of 0.39, with a Pearson correlation of 0.97. A large part of any SSR is therefore the Sharpe ratio in another unit.

It also depends on the length of the history. Under independent monthly returns with an annualised Sharpe ratio of 1.0, the mean SSR is 0.80 at the 120-month reporting gate, 0.50 at 220 months (the shortest history in the sample), 0.39 at 499 months (the median) and 0.36 at 1,519 months (the longest). Kernel estimates of a long-run variance are biased downward when the number of rolling estimates is not large relative to the window, so shorter histories get higher SSRs for the same Sharpe ratio. SSR levels computed over histories of very different length are therefore not directly comparable, even with the same window and bandwidth. In this sample the effect on rankings is moderate, because most histories run between 439 and 630 months.

The multiple removes the Sharpe level. The permutation multiple has a median of 1.12 and an interquartile range of 0.99 to 1.22; it exceeds one for 127 of the 173 variants. A multiple above one means that the rolling Sharpe ratio was steadier, relative to its mean, than the same returns in random order would make it; below one, that good and bad months were more clustered in time than chance ordering produces. Its rank correlation with the Sharpe ratio is −0.01, against 0.64 for the SSR.

No single multiple is established. Twenty-five variants have a permutation p-value below 0.05, against about 8.7 expected if no series had any order in time and the tests were independent. The tests are not independent, since variants of a family share returns, and no variant remains below the threshold after the Benjamini and Hochberg (1995) false discovery rate adjustment at 5 percent. The excess of small p-values across the catalog is suggestive; no individual variant's multiple is established.

Table 4 continues the worked example. The Triplet's SSR of 0.94 is 2.00 times the 0.47 that its own returns give in random order. HAA's SSR of 0.48 is below the 0.57 of its own returns in

random order, a multiple of 0.84: the long waves of Figure 1 are more clustered than chance. The full table and the distribution of multiples are in Appendix D.7.

Table 4. Random-order benchmark for selected series: full-history Sharpe ratio and SSR, random-order benchmark (Random), permutation multiple and p-value; the last column is the multiple on the window from January 2000.

Series	Sharpe	SSR	Random	Multiple	p	2000 on
Momentum-Corr. Triplet	1.21	0.94	0.47	2.00	0.002	1.90
DMS Triad	1.24	0.70	0.47	1.49	0.027	1.44
Pragmatic AA	1.18	0.73	0.45	1.63	0.012	1.42
HAA, Hybrid AA	1.50	0.48	0.57	0.84	0.796	1.70
GEM	1.10	0.42	0.42	1.00	0.447	1.09
Classic 60/40	0.79	0.31	0.29	1.05	0.296	1.14
S&P 500 (VFINX)	0.82	0.32	0.31	1.02	0.407	1.13
60/40 (VFINX/VBMFX)	0.96	0.34	0.36	0.94	0.563	1.13

5.3 The ranking depends on the decade, not on the estimator settings

Settings. Changing the window (24 to 60 months) or the bandwidth (5 to 71 lags) moves levels but hardly moves rankings: the rank correlations with the baseline specification run from 0.92 to 0.99. The median SSR is 0.85 with 5 lags, 0.56 with 15, 0.45 with 35 and 0.42 with 71. Individual variants can still move: one moves 55 places between the 35- and 71-lag specifications (Appendix E.1, Table E.1).

Decades. To see whether the ordering persists, we recompute the SSR inside four calendar subperiods for every published variant with at least 72 monthly returns in them (Table 5). Between adjacent subperiods the rank correlation of the cross-section is 0.08, −0.38 and 0.11. Higher SSRs in the 2000s go with lower SSRs in the 2010s: an imperfect association rather than a complete reversal, but not an artefact of the bandwidth (−0.37 to −0.28 under the other bandwidths). The benchmarks move with it. The S&P 500 goes from 0.15 in the 2000s to 1.63 in the 2010s and the 60/40 blend from 0.23 to 1.60, while the median static variant goes from 0.35 to 1.58 and the median tactical variant hardly moves (0.89 and 0.88); from 2020 the ordering of the two types reverses again (1.01 against 0.61). Figure 3 shows the S&P 500's path: its rolling Sharpe ratio spent 2002 to 2004 and late 2008 to 2010 below zero, then held high for a decade.

Table 5. SSR by subperiod: number of variants, medians (72-month minimum, so levels are not comparable with the full-history values), two benchmarks, and the rank correlation of the cross-section with the previous subperiod (pairwise n in brackets). Full table in Appendix D.6.

	Before 2000	2000 to 2009	2010 to 2019	2020 to Aug 2026
Variants with an estimate	161	172	173	174
Median SSR, tactical	0.63	0.89	0.88	1.01
Median SSR, static	0.68	0.35	1.58	0.61

	Before 2000	2000 to 2009	2010 to 2019	2020 to Aug 2026
S&P 500 (VFINX)	0.52	0.15	1.63	0.81
60/40 (VFINX/VBMFX)	0.74	0.23	1.60	0.59
Rank correlation with previous subperiod		0.08 (161)	−0.38 (172)	0.11 (173)

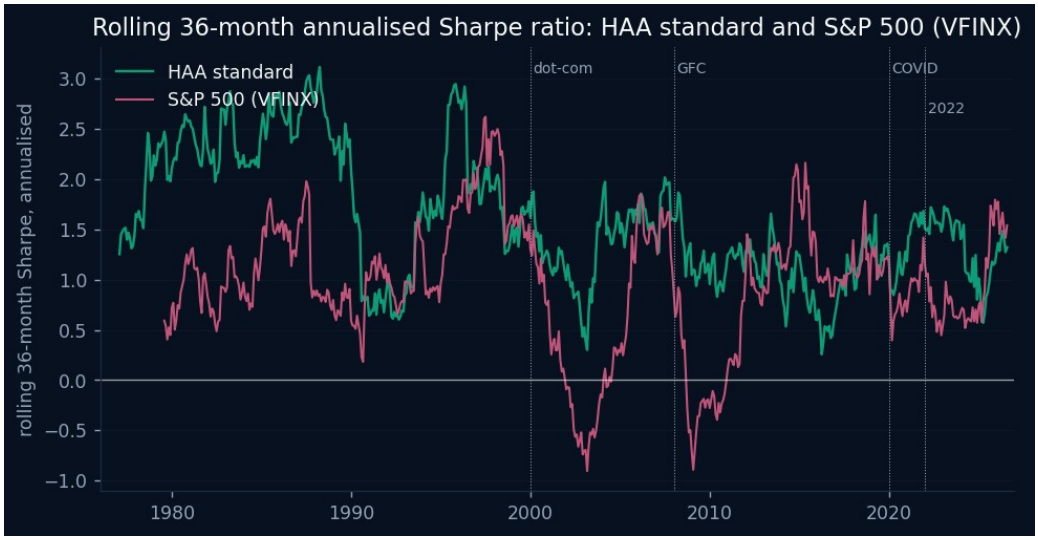


Figure 3. Rolling 36-month annualised Sharpe ratio of Hybrid Asset Allocation (rolling series from February 1977) and the S&P 500 proxy VFINX (from August 1979), through August 2026.

The multiple is period dependent too. On the window from January 2000 the median permutation multiple is 1.24 and its rank correlation with the full-history multiple is 0.59. HAA's multiple moves from 0.84 to 1.70 (Table 4), approaching the Triplet's 1.90, while the DMS Triad stays close to its full-history value. Whether a strategy's rolling Sharpe ratio looks more or less orderly than chance depends on which decades the window holds.

The design does not separate the possible explanations: the finite-sample variability of the denominator when a period holds few rolling estimates (at most 45 in the last subperiod, against a 35-lag bandwidth), the loss of between-regime variation in short periods, the restart of the rolling windows at each boundary, unequal history lengths within a period, and genuine changes in how strategies behaved. Appendix D.6 discusses them.

5.4 Which strategies score high

The top of the ranking is dominated by monthly momentum rotations with diversified defensive legs: the Momentum-Correlation Triplet family (0.86 to 0.94), Pragmatic Asset Allocation (0.73), the DMS Triad family (0.66 to 0.72), two Vitral multi-asset momentum variants (0.66 and 0.67) and the KISS momentum blends (0.58 to 0.65). The bottom holds single-asset trend rules on bonds and gold (0.11 to 0.18), daily volatility-regime strategies (0.20 to 0.22), a gold-based cross-asset rule at the catalog minimum (0.09) and static allocations (Table D.3). By group, tactical variants have a higher median SSR than static ones (0.46 against 0.39), and monthly-rebalanced strategies (0.47) a higher one than daily (0.36) or annually rebalanced

ones (0.36) (Table D.2). Static allocations are not uniformly low: the Golden Butterfly (0.49) and All Weather (0.51) sit above the tactical median.

Turnover is positively associated with the SSR: the rank correlation is 0.26 across the 173 variants, 0.19 within tactical variants and 0.14 within monthly tactical variants, and the median SSR rises from 0.42 in the lowest turnover tercile to 0.48 in the highest (Table D.8, Figure D.3). These comparisons do not isolate a turnover effect: strategy type, rebalancing frequency, leverage and asset exposures all move with turnover, and all 31 static variants sit in the lowest tercile.

5.5 Uncertainty

A moving block bootstrap (36-month blocks, 2,000 replicates) gives an idea of how much each SSR could vary. All but one of the 173 percentile intervals exclude zero (the exception, the trend rule on gold, reaches −0.01), no variant's 5th percentile exceeds 0.5, 10 variants have intervals entirely below 0.5, and the median interval width is 0.44; the Triplet's interval is [0.43, 1.02]. An interval above zero concerns positive mean rolling performance; it does not show that a series is steadier than chance, which is the question of Section 5.2. For the steadiest strategies the bootstrap distributions are not centred on the estimates (the Triplet's bootstrap mean is 0.67), so the intervals are descriptive and no coverage is claimed (Appendix E.2).

5.6 Above cash: what survives

The SSR in this paper uses a cash reference of zero, as its authors recommend, which also keeps the results comparable with theirs. Investors usually judge a strategy against cash, and the platform whose catalog is analysed here publishes its Sharpe ratio and its Stability value above cash. We therefore repeat the headline results with every monthly return measured above US dollar cash,

$$\tilde{r}_t = r_t - c_t \tag{10}$$

where c_t is the return of the platform's cash index (a Treasury bill fund chain, from 1900) between the same two month-ends as r_t. Equations (1) to (9) are applied to $\tilde{r}_t$ in place of r_t. The sample, the window, the bandwidth, the reporting gate and the bootstrap and permutation seeds are unchanged, and the Deflated Sharpe Ratio uses the platform's stored above-cash moments with the same 219 trials (no-skill benchmark 0.46, annualised). Recomputed from the month-end series, the above-cash monthly Sharpe ratios of all 219 trial variants agree with the values the platform stores to the sixth decimal. Table 6 sets the two references side by side, Figure 4 compares the two SSRs variant by variant, and Appendix E.3 gives the full results.

Table 6. Headline results with a cash reference of zero (Sections 5.1 to 5.5) and above US dollar cash, 173 published variants, full history through August 2026. DSR: Deflated Sharpe Ratio with 219 trials, on zero-reference and above-cash moments respectively.

	Zero reference	Above cash
SSR: minimum / median / maximum	0.09 / 0.45 / 0.94	0.01 / 0.32 / 0.78
Highest SSR	Triplet, standard	Low-Initiative LETF v2

	Zero reference	Above cash
Rank correlation of the SSR with the Sharpe ratio / DSR	0.64 / 0.59	0.77 / 0.73
Rank correlation with growth rate / drawdown depth	0.15 / 0.54	0.48 / 0.23
DSR at or above 0.90	146	112
of which SSR at or above / below the cutoff	53 / 93 (cutoff 0.5)	46 / 66 (cutoff 0.37)
Windows and bandwidths: rank correlation with the baseline	0.92 to 0.99	0.92 to 0.99
Adjacent subperiods: rank correlations	0.08 / −0.38 / 0.11	−0.18 / −0.37 / 0.26
S&P 500: 2000s / 2010s	0.15 / 1.63	0.08 / 1.41
S&P 500 / 60/40, full history (catalog 25th percentile)	0.32 / 0.34 (0.37)	0.22 / 0.25 (0.26)
Random-order benchmark: slope on the Sharpe ratio	0.39	0.40
Permutation multiple: median; rank correlation with the Sharpe ratio	1.12; −0.01	1.08; 0.22
Multiples above one; p-values below 0.05 (8.7 expected); surviving the adjustment	127; 25; 0	115; 14; 0
Median SSR, tactical / static	0.46 / 0.39	0.33 / 0.28
Rank correlation with turnover	0.26	0.35
Bootstrap: intervals above zero; 5th percentiles above 0.5	172; 0	170; 0

Levels fall, most for long histories. Subtracting cash lowers each rolling Sharpe ratio by roughly the cash return of its window over its volatility, so the levels fall: the median SSR goes from 0.45 to 0.32, and the ratio of the two values has a median of 0.75 across the variants. The fall is largest for histories that hold the high-rate 1970s and 1980s (a median ratio of 0.72 for the 148 histories that start before 1990, 0.80 for the 25 that start later and 0.95 for the seven that start in 2000 or later). Allocations with large bond, gold or cash weights over those decades fall furthest: the static Permanent Portfolio, All Weather and a trend-tilted risk-parity blend, with histories from 1960 and 1972, move from 58th, 49th and 23rd to 162nd, 152nd and 163rd. With 0.5 as the cutoff only 17 variants would pass, so above cash we use 0.37, the split the platform adopted for its Stability flag because it marks the same share of its published variants as 0.5 did with a zero reference. In this sample 0.37 flags 122 variants, against 118 below 0.5 with a zero reference.

The ranking keeps its broad shape. The rank correlation between the two SSRs is 0.71, about the same as between the two Sharpe ratios (0.73). At the two cutoffs (0.5 with the zero reference, 0.37 above cash) 139 of the 173 variants, 80 percent, fall on the same side: 103 are low under both and 36 high under both. Five of the twelve highest and nine of the twelve lowest are the same under both references, and the median variant moves 19 places. The Triplet family

stays near the top (the standard variant is second, at 0.63), the trend rules on gold and long Treasuries and the gold-based cross-asset rule stay at the bottom, and the S&P 500 and the 60/40 blend stay below the catalog's lower quartile. The leading names otherwise change: Low-Initiative LETF v2, a daily leveraged rotation whose history starts in 2002 and so holds no high-rate decade, comes first (0.78), and five HAA variants enter the top twelve (Table E.2).

The worked example changes in size, not in direction. Above cash, HAA has a Sharpe ratio of 1.10 and an SSR of 0.56, ninth in the catalog, and the Triplet 0.96 and 0.63, second. HAA's annualised rolling Sharpe ratio averaged 2.34 from 1977 to 1989 and between 1.05 and 1.57 in each later decade with a zero reference; above cash the same averages are 1.47 and between 1.01 and 1.09. Subtracting cash removes most where HAA's rolling Sharpe ratio was highest, in its high-rate first decade, and so flattens the long wave of Figure 1: its long-run standard deviation falls to 4.3 times its ordinary one, and its permutation multiple rises from 0.84 to 1.32 ($p = 0.059$). The Triplet still has the higher SSR; the gap between the two is much smaller.

The four findings hold. First, the SSR still ranks the catalog differently from the Sharpe ratio and the Deflated Sharpe Ratio (rank correlations 0.77 and 0.73, higher than with the zero reference), and most variants the Deflated Sharpe Ratio calls credible are unsteady: of the 112 with a score of at least 0.90, 66 fall below 0.37 (and 96 below 0.5). Two relations change: above cash the SSR is moderately related to the growth rate (0.48) and only weakly to the depth of the worst drawdown (0.23), the reverse of the zero reference, and far fewer Deflated Sharpe Ratios round to 1.00 (29 rather than 89), so the tie-breaking role of Section 5.1 matters less. Second, the random-order benchmark is again about 0.4 times the Sharpe ratio and no permutation multiple survives the false discovery rate adjustment, but the multiple is now weakly related to the Sharpe ratio (0.22), and 14 p-values fall below 0.05 against 8.7 expected, so the excess of small p-values is weaker than with the zero reference. Third, rankings are again stable across windows and bandwidths and unstable across decades: the 2000s-to-2010s rank correlation is −0.37 (−0.34 to −0.29 under the other bandwidths), the S&P 500 goes from 0.08 in the 2000s to 1.41 in the 2010s, and the median static variant from 0.24 to 1.35, although above cash the median tactical variant rises too (from 0.64 to 0.85). Fourth, tactical variants still score higher than static ones (0.33 against 0.28) and monthly rebalancing higher than daily or annual (0.33 against 0.29 and 0.27), and turnover is more strongly associated with the SSR (0.35; 0.31 within tactical variants).



Figure 4. Full-history SSR above US dollar cash against the zero-reference SSR, 173 published variants, with the S&P 500 and the 60/40 blend, the 0.5 cutoff of the zero-reference SSR and the 0.37 split used above cash. The dashed line marks equal values.

In short, the SSR's ranking and its flags survive the change of reference in broad terms, with a rank correlation of 0.71 and 80 percent of the flags unchanged, and the paper's message holds under both references. Individual placings do depend on the reference, mostly through how much of a history lies in high-rate decades, so an SSR, like a Sharpe ratio, should be quoted with its cash reference.

6. Limitations

- **Backtests, not track records.** Every strategy series is a backtest produced by one engine under one execution convention, and the benchmark series are built from observed fund prices. The SSR inherits whatever look-ahead or proxy error the underlying series carry.
- **A selected catalog.** Strategies entered the catalog because their published or proposed backtests were attractive. Its Deflated Sharpe Ratio is a partial model of the recorded search and does not correct for catalog selection, for the searches upstream in the published research (Bailey, Borwein, López de Prado and Zhu, 2014; Harvey, Liu and Zhu, 2016), or for data errors.
- **Not independent strategies.** Variants within a family are correlated and two share a return path, so cross-sectional counts overstate the number of independent strategies.
- **Cash reference.** The main results use a cash reference of zero. Above cash (Section 5.6) the rolling Sharpe ratios of the high-rate decades fall, levels drop by about a

quarter, long histories through those decades fall most, and the ordering changes in its details (rank correlation 0.71) while the four findings hold. Neither reference is the single right one: zero is the source paper's recommendation, cash is the investor's opportunity cost.

- **Two tuning choices.** The bandwidth moves levels by a factor of two while leaving aggregate orderings largely intact, and the fixed 35-lag specification is a descriptive analogue rather than a consistent estimate of the structural ratio (Appendix A.1). Levels here should not be compared with levels computed under an automatic bandwidth rule, nor, without the benchmark of Section 5.2, across histories of very different length.
- **Bootstrap.** It uses fixed block lengths, its distributions are biased for the steadiest strategies, and no coverage has been validated.
- **Permutation benchmark.** It conditions on each series' realised returns and treats them as exchangeable under its null; it says nothing about whether those returns would recur, and its p-values are marginal, one per series, in a catalog whose variants share returns.
- **Subperiods.** The design compares unequal periods and cohorts, and the negative correlation between the 2000s and the 2010s, although robust to the bandwidth, has not been separated from estimation effects.
- **No prospective test.** The comparisons show no consistent positive rank persistence across the selected historical periods, and prospective predictive value remains untested. Such a test does not require live histories: it requires frozen rules, defensible publication dates, point-in-time inputs and genuinely withheld evaluation data, in the sense of the discovery and embargo samples of López de Prado, Lipton and Zoonekynd (2026), and the retrospective selection of the present catalog makes those safeguards necessary.

7. Relation to the literature

The statistics that dominate backtest evaluation describe the full-sample Sharpe ratio: its sampling distribution (Lo, 2002; Mertens, 2002; Opdyke, 2007; Ledoit and Wolf, 2008; López de Prado, Lipton and Zoonekynd, 2026), the probability that it exceeds a benchmark (the Probabilistic Sharpe Ratio) and the correction for the number of trials (the Deflated Sharpe Ratio). In the terminology of Bajo Traver and Rodriguez Dominguez (2026) they are point-in-time. The SSR instead treats the rolling Sharpe ratio as a time series, studentises its mean with a HAC standard deviation (Newey and West, 1987) and recommends a moving block bootstrap (Künsch, 1989) for inference. Its authors apply it to seventeen hedge fund style indices over 29 years and find SSRs of 0.27 to 0.62, rankings that survive changes of window and bandwidth while levels do not, and an average rank correlation of 0.31 across three subperiods. This catalog agrees on the first two points; across its subperiods the rank correlations are lower (0.08, −0.38 and 0.11), although samples, estimators and estimation uncertainty differ too much for a close comparison. Appendix F gives the full review, including a check of the autocorrelation-adjusted Sharpe ratio variance on these series.

8. Conclusion

Applied to 173 asset allocation backtests, the SSR does what it promises: it separates strategies with similar Sharpe and Deflated Sharpe ratios by how steadily they earned them, and it resolves ties among the many variants whose Deflated Sharpe Ratio rounds to one (fewer above cash, where the Deflated Sharpe Ratio saturates less). Its aggregate rankings are stable across the tested windows and bandwidths, although individual ranks can move substantially. Its most important property in this sample is period dependence: the relative SSR of tactical strategies, static allocations and equity benchmarks changes from one decade to the next, and the S&P 500 moves from among the least stable series of the 2000s to among the most stable of the 2010s. That need not be a flaw in the statistic: one reading is that it reports, in one number, how different a strategy's risk-adjusted path looks depending on which regimes the window contains, although Section 5.3 lists the estimation effects that have not been separated from that reading. The permutation benchmark separates the two things the statistic combines: a level that scales with the Sharpe ratio and the length of the history, and an ordering component that is itself period dependent. Measured above cash, levels fall and the leading names change, but these conclusions do not.

For users of the statistic this means four things: state the window, the bandwidth, the history length and the cash reference with every SSR; compare SSRs over common calendar dates; divide by the random-order benchmark before comparing histories of very different length; and read the SSR, as its authors suggest, as a description of a track record rather than a forecast. Interpretation remains conditional on data provenance, sample selection and the estimator's finite-sample behaviour.

Disclosure

The author operates BestFolio, the platform whose catalog is analysed here, and has a commercial interest in it. The study used the platform's stored series and public API and received no external funding. The statistic, its implementation and the data files are described in Appendices A and B so that the results can be checked independently.

AI use

The author used generative AI tools to assist with parts of the analysis code and the drafting of this paper, reviewed and edited the output as needed, and takes full responsibility for the content.

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Appendix A. The estimator in full

A.1 Estimator

For each series we compute monthly simple returns $r_1, \ldots, r_T$ from completed month-end net asset values. With window $\tau = 36$ the rolling per-period Sharpe ratio is

$$Z_t = \frac{\bar{r}_{t,\tau}}{\hat{\sigma}_{t,\tau}}, \hat{\sigma}^2_{t,\tau} = \frac{1}{\tau - 1} \sum_{j=0}^{\tau-1} \left(r_{t-j} - \bar{r}_{t,\tau} \right)^2, t = \tau, \ldots, T \tag{A.1}$$

with the zero cash reference of Appendix B.2 (above cash in Section 5.6 and Appendix E.3, where r_t is replaced by $r_t - c_t$) and no annualisation, which cancels in the ratio. Reindex the rolling process as $Z_1, \dots, Z_N$ with $N = T - \tau + 1$, let $\overline{Z}$ be its mean, and let

$$\hat{\gamma}_k = \frac{1}{N} \sum_{t=k+1}^{N} (Z_t - \overline{Z})(Z_{t-k} - \overline{Z}), \hat{\sigma}_L^2 = \hat{\gamma}_0 + 2 \sum_{k=1}^{L} \left(1 - \frac{k}{L+1}\right) \hat{\gamma}_k \tag{A.2}$$

be the sample autocovariances with divisor N and the Bartlett-kernel long-run variance with bandwidth L. The reported statistic is $\widehat{SSR} = \overline{Z} / \hat{\sigma}_L$ with benchmark $\mathrm{SR}^\star = 0$ and $L = \tau - 1 = 35$. When a series has fewer rolling estimates than $L+1$, the bandwidth is reduced to $N - 1$; this never binds in the full-history sample, where $N \geq 185$, but does in some subperiod calculations (Appendix D.6). Throughout, K denotes trials, T monthly returns, N rolling estimates and ℓ the bootstrap block length.

The statistic is a fixed-bandwidth descriptive analogue of the structural SSR rather than a consistent estimate of it. A fixed bandwidth does not in general estimate the infinite-horizon variance consistently, and the Bartlett weights recover only part of the overlap-induced dependence even when the bandwidth spans it: under the mechanical-overlap approximation of Proposition 3.1 with $\tau = 36$, the long-run variance ratio $\sigma_\infty^2 / \gamma_0 \approx 1 + 2 \sum_{k=1}^{35} (1 - k/36) = 36$, whereas the Bartlett estimate with $L = 35$ approaches $1 + 2 \sum_{k=1}^{35} (1 - k/36)^2 \approx 24$, so the fixed-bandwidth SSR sits roughly 22 percent above the structural target under that approximation. We therefore do not apply the source paper's standard-normal calibration to this specification, and we present the statistic as a ranking aid whose levels are specific to the window and bandwidth stated and, as Appendix D.7 shows, to the length of the history.

The statistic is reported as missing, by policy rather than mathematical necessity, when fewer than 120 monthly returns are available; it is undefined when any rolling window has a standard deviation at or below 10^{-12} or when the long-run variance is not positive. At the gate $N = 85$ and $L/N \approx 0.41$; the ten-year threshold is a reporting choice and has not been calibrated in finite samples, although under independent returns with a Sharpe ratio of 1.0 the statistic's mean at the gate is about twice its mean at the sample's median history (Appendix D.7). In the primary sample the shortest history is 220 months, so the results say little about behaviour near the gate.

A.2 Bandwidth

The source paper selects the bandwidth with the Andrews (1991) automatic procedure, which returned $L = 15$ on all seventeen of its 29-year monthly series, and reports that the sample-size formula $\lfloor 4 (N/100)^{2/9} \rfloor$ returns $L = 5$ on the same data, with SSR levels 34 percent lower under Andrews and a rank correlation of 0.98 between the two. We have not implemented Andrews' procedure. For a comparative table we prefer one fixed specification for all series, $L = \tau - 1 = 35$, the full extent of the mechanical overlap, so that every strategy is measured by the same estimator and the levels are reproducible from the definition alone. Appendix E.1 reports the sensitivity to $L \in \{5, 15, 35, 71\}$ and to the same sample-size formula, which selects 4 to 7 lags

on these series (3, 157, 12 and 1 variants respectively); we call it the sample-size rule to distinguish it from the full data-dependent Newey and West (1994) procedure. The fixed 35-lag specification produces lower point estimates than the fixed 5- and 15-lag specifications and the sample-size rule; we have not established conservative inferential coverage for it, and increasing the bandwidth further does not lower every variant's value.

A.3 Bootstrap

We follow the source paper's recommendation and use a moving block bootstrap (Künsch, 1989): contiguous blocks of ℓ monthly returns are drawn with replacement from the original series and concatenated to its length, the rolling process and the fixed-bandwidth SSR are recomputed on each replicate, and $B = 2{,}000$ replicates are summarised. Rather than the automatic block length of Politis and White (2004), as corrected by Patton, Politis and White (2009), we report two fixed lengths, $\ell = 12$ and $\ell = 36$ months; results are reported for $\ell = 36$ and checked against $\ell = 12$. Resampling return blocks and recomputing the rolling statistic recreates the mechanical overlap dependence, but a block equal to the window length does not preserve the original regime structure, and we make no claim that it does.

For each variant we report the 2.5th and 97.5th percentiles of the replicates (the percentile interval), the basic interval $[\,2\widehat{SSR} - q_{0.975}, 2\widehat{SSR} - q_{0.025}\,]$, and the uncentered bootstrap proportion $q_{\le c} = B^{-1}\sum_{b} 1\{\widehat{SSR}^{\star(b)} \le c\}$ for $c = 0.5$. The proportion is not a posterior probability and is not generated under a null; its only inferential reading is through the one-sided percentile bound, rejecting $H_0 : \mathrm{SSR} \le c$ when the 5th bootstrap percentile exceeds c, subject to the method's coverage, which we have not validated. The Monte Carlo standard error of a tail proportion of 0.05 with $B = 2{,}000$ is about 0.005.

A.4 Companion statistics and verification

For every variant we also compute the platform's Sharpe ratio (annualised mean over annualised sample deviation of the same monthly returns, zero cash reference), the Probabilistic Sharpe Ratio at benchmark zero from the same monthly moments, the Credibility-Adjusted SSR of the source paper, and, for the principal comparisons, the components $\overline{Z}$, the naive standard deviation of Z and its HAC standard deviation. Turnover, strategy type, rebalancing frequency, leverage and risk category are taken from the leaderboard payload; the Deflated Sharpe Ratio is recomputed with 219 trials (Appendix C.2).

The statistic was implemented twice: in a standalone script written from the definitions above, and in the platform's production service, written independently from a specification of the same definitions and tested against an independently computed reference fixture. On identical inputs (the series of 17 October 2026 with the partial month, as stored by the platform) the two agree to 15 decimals on 217 of the 219 active variants, the remaining two being below the production gate; the check is retained with the replication package. The standalone script also reproduces the Deflated Sharpe Ratios of Appendix C.2 to their sixth decimal.

Appendix B. Data, sample and replication

B.1 The catalog and the sample

BestFolio maintains a catalog of rules-based asset allocation strategies, each with one or more parameter variants, and publishes for each variant a full-history backtest, periodic signals and summary statistics. The sample is defined on the catalog as it stood on 5 September 2026 (production release release/2026-09-05-8b44fa1): 219 active variants, of which 177 were published on the public leaderboard and 42 were research variants that were tested but not published at that date. The series used here are those of the same 219 variants as re-run by the platform on 17 October 2026 (code 896bf8a, methodology drift-until-flip-v1), after the restatement of 28 September 2026 (release release/2026-09-28-e9a2a80) and an audit of the proxy chains; Appendix B.2 describes the resulting conventions, and Appendix G the changes. All 219 enter the Deflated Sharpe Ratio's count of trials.

The primary sample is the published set less two restrictions. First, three variants of one strategy whose macroeconomic input is known to have been reconstructed with revised rather than point-in-time data are excluded pending restatement; the supplementary file identifies them. Second, the SSR is reported only for variants with at least 120 completed monthly returns, which excludes one further variant with 83 months. The primary sample therefore holds 173 variants from 92 strategy families. Appendix E.1 reports the sensitivity of the headline statistics to the first restriction: including the three excluded variants changes the median SSR by less than 0.01 and the adjacent-decade rank correlations by less than 0.01.

The strategies fall into two types. Tactical variants (142) change their allocation on a schedule according to momentum, trend, breadth, volatility or macroeconomic rules; they include reconstructions of the Global Tactical Asset Allocation family (Faber, 2007), Global Equities Momentum (Antonacci, 2014), and the Protective, Vigilant, Defensive, Bold and Hybrid Asset Allocation family (Keller and Keuning, 2016, 2017, 2018, 2022, 2023), as well as house strategies and community-contributed variants. Static variants (31) are fixed-weight allocations rebalanced on a schedule, from the classic 60/40 to the Permanent Portfolio, the Golden Butterfly and All Weather. By rebalancing frequency there are 121 monthly, 24 daily, 17 annual, 10 quarterly and 1 weekly variants; 52 use leveraged exchange-traded funds or synthetic leverage. Variants of one strategy share rules and often assets, so the 173 series are not independent draws; two variants (a US-equity-only allocation listed under two strategy names) have practically identical monthly returns over their common history from February 1985 (largest difference 0.007 percentage points), although one of them now starts in 1900. We report the catalog as published and note where family structure matters.

B.2 Series construction and conventions

Every backtest is simulated daily by the platform's engine on adjusted prices. The platform's published methodology states one-way transaction costs of 10 basis points and execution on the session after a signal; the study uses the stored net asset value series and does not re-run the engine. Where an instrument's history is shorter than the backtest, the engine substitutes documented proxies (index funds, futures-based series, or synthetic leverage built from an index), so the earliest histories reach back to 1900. Fund histories used as proxies are total return: distributions missing from the primary price provider's records, mostly before 1990, are

restored from a second provider and checked against the funds' published returns, and the S&P 500 proxy until early January 1987 is the index's daily total return. When a strategy's rules set an earliest permitted start, its indicators are computed on the price history available before that date and its backtest begins at that date. Between signals the engine holds the positions bought at the last signal and lets their weights drift with prices; at a signal it trades from the drifted weights to the new target weights, and the one-way turnover reported for a variant is half the sum of absolute differences between the drifted and the target weights at each signal, summed over a year and averaged over the backtest. Daily strategies emit a signal only when their regime or allocation changes, so single-asset regimes trade only when their signal flips; static allocations trade back to their target weights on their rebalancing schedule, and monthly rotations at most once a month.

The stability statistic is computed on monthly returns from completed month-end observations. The series end on 16 October 2026; to keep the dated cut we use returns through 31 August 2026 and drop the later observations (September 2026 and the partial October). The platform's strategy pages compute their Stability value on the whole stored history including the current partial month, and above cash (Section 5.6). Keeping the later observations changes little: with a zero cash reference the rank correlation between the two treatments across the 173 variants is 0.9998, the largest difference in any variant's SSR is 0.020, and the median is 0.447 with completed months through August and 0.448 with the later observations; above cash the same figures are 0.9998, 0.014, 0.319 and 0.320. Of the 173 variants, 35 start before 1980, 148 before 1990, 166 before 2000 and 172 before 2008; the median history is 499 months and the longest 1,519.

Returns are used with a cash reference of zero, as the source paper recommends: the rolling Sharpe ratio is the mean of raw monthly returns over their standard deviation, not of excess returns over a cash rate. Across a century of changing short rates this raises Sharpe levels in high-rate decades relative to an excess-return definition; every series is treated the same way, Section 5.6 and Appendix E.3 repeat the results above cash, and Section 6 lists the choice as a limitation. The monthly moments used for the Probabilistic Sharpe Ratio are recomputed from the same returns; the stored monthly Sharpe ratios on the platform agree with the recomputed ones to within 0.008 per month, the difference coming from the observations after August 2026 that they include. The compound growth rate, maximum drawdown and turnover are imported from the platform's leaderboard payload as published on 17 October 2026 and are not recomputed; the Deflated Sharpe Ratio is computed as described in Appendix C.2.

Benchmarks are constructed from the same price database: the Vanguard 500 Index Fund (VFINX, from August 1976) as the S&P 500 proxy, the Vanguard Total Bond Market Index Fund (VBMFX, from December 1986) for US bonds, a monthly-rebalanced 60/40 blend of the two, and the gold (GLD, from November 2004), long Treasury (TLT from July 2002, VUSTX from May 1986) and Nasdaq 100 (QQQ, from March 1999) funds. The month-end closes are those of the 5 September 2026 extract with the same total-return corrections as the strategy series: until 5 January 1987 the S&P 500 series is the index's daily total return, which supplies the distributions and capital gains missing from the VFINX record, and the distributions missing from the VBMFX and VUSTX records before 1990 are restored. Their windows are stated with every number.

B.3 Replication and provenance

Data. The sample is defined on the catalog of 5 September 2026 (release release/2026-09-05-8b44fa1): 219 active variants, 177 of them published. The month-end net asset value series of the same 219 variants, and the stored monthly moments of each variant's latest backtest (zero-reference and above-cash), were extracted read-only from BestFolio's production database on 17 October 2026, after that day's re-run (code 896bf8a, methodology drift-until-flip-v1), together with the platform's daily US dollar cash index (a Treasury bill fund chain from 1900) used in Section 5.6. The benchmark month-ends are the adjusted closes extracted on 5 September 2026, with the restatement's total-return corrections applied to VFINX, VBMFX and VUSTX by the platform's own correction code (build_bench_monthend.py). The leaderboard fields (turnover, strategy type, frequency, leverage, risk category, compound growth rate, maximum drawdown) were taken from the public endpoint /api/performance/leaderboard on 17 October 2026 for the same 177 published variants, and the Deflated Sharpe Ratio at $K = 219$ was computed with the platform's own function on the stored moments of the 219 latest runs (build_inputs.py). The public scores of that day could not serve as a check: they came from a cache built before the platform's last runs of the day. The study script that produced the numbers in Sections 3 to 5 and Appendices A to E and the tables (ssr_study_v07.py, Python with NumPy, which is ssr_paper_study_v2.py with an option to measure returns above cash), the chart script, the independently computed reference fixture used to test the production implementation, the production-agreement check, the study outputs and the supplementary tables are published at https://github.com/lbonherbe/sharpe-stability-ratio-taa. The frozen month-end extracts are retained by the author and available to referees on request; they are not redistributed because the benchmark prices are licensed data and the catalog extract includes the unpublished research variants. The autocorrelation-adjusted Deflated Sharpe check of Appendix F.1 and the reproduction of the Deflated Sharpe Ratios in Appendix A.4 come from a separate script (ssr_catalog.py) run on the extract of 17 October 2026 including the observations after August; its output is retained with the replication files. The permutation benchmark of Appendix D.7 comes from a third script (ssr_permutation_v07.py) run on the study's data with the same estimator; it stops unless every real SSR it computes equals the supplementary table, and its per-series output (permutation_results.csv), aggregate numbers (permutation_results.json) and console log are in the archive. The supplementary variant table holds the 177 published variants, with a flag on the three revised-data variants (the variant below the ten-year gate carries empty SSR fields) and, for each, the Sharpe ratio, Deflated Sharpe Ratio, Probabilistic Sharpe Ratio, the SSR on the full history and on the 2000 and 2008 windows, the four subperiods with their return counts, the four joint window-bandwidth specifications, the four fixed bandwidths and the sample-size rule, the annualised mean rolling Sharpe ratio with the naive and HAC standard deviations of the rolling series, and, for the 173 eligible variants, the bootstrap mean, percentile and basic intervals, 5th percentile and proportion at or below 0.5 for 36-month blocks and the percentile interval and proportion for 12-month blocks.

Platform. The full-history statistic in US dollars is served on each strategy's summary endpoint, including the current partial month by the platform's convention and measured above cash; the above-cash values of Section 5.6 correspond to it with completed months through August 2026 only. The public leaderboard endpoint (/api/performance/leaderboard, field sharpe_stability, metadata ssr_meta with definition identifier ssr-36m-nw-excess-v2 since 18 October 2026, and ssr-36m-nw-v1, the zero-reference definition used in this paper, before that) also carries the field

for every leaderboard period and currency. The platform's methodology page documents the backtest conventions (transaction costs, execution timing, proxy chains) and the definition of the published Sharpe ratio. The platform's Stability flag uses a split of 0.37 above cash, calibrated to flag the same share of its published variants as 0.5 did with a zero reference.

Appendix C. The Deflated Sharpe Ratio and the count of trials

C.1 Why K = 219

Every number in this paper refers to one dated cut: the catalog of 5 September 2026, which defines the 219 active variants, the 177 published ones and the 173 eligible ones, evaluated on the series as re-run on 17 October 2026 with completed months through August 2026. The count of trials for the Deflated Sharpe Ratio belongs to the same cut: $K = 219$ is the number of variants that had been tried, published or not, when the sample was fixed. The platform's live score is a moving figure. It counts every variant with a stored backtest, active or retired, as a trial, except, since 18 October 2026, single-fund holdings that apply no rule (eight in the database of 17 October 2026, one of them a cash fund whose zero-reference Sharpe ratio would otherwise inflate the benchmark): applied to that database, the rule gives 278 trials, and the count will keep growing as strategies are added. Adopting it would tie one column of the paper to a later and changing catalog while every other number stays on the 5 September cut. We therefore keep $K = 219$, and report the live-count values beside it so that a reader can switch: with 278 trials the no-skill benchmark is 0.60 rather than 0.58, the scores differ by 0.001 at the median and by at most 0.061 (the all-equity US allocation, whose history now starts in 1900), the variants at or above 0.90 fall from 146 to 143, and the cells of Table 3 become 53, 90, 2 and 28. On the above-cash moments of Section 5.6 the same live universe gives a benchmark of 0.52 (0.46 with $K = 219$). Since 18 October 2026 the platform publishes its score on those above-cash moments, so its live figures differ from the zero-reference ones used here.

C.2 The platform's Deflated Sharpe Ratio

The Deflated Sharpe Ratio used here is computed with the platform's own implementation on the paper's trial set: the 219 variants of Appendix B.1 count as trials ($K = 219$), the cross-trial variance of monthly Sharpe ratios is taken from that same set, and each variant's own history length and monthly moments enter the Bailey and López de Prado (2014) formula. The resulting no-skill benchmark is an annualised Sharpe ratio of 0.58. Under the platform's current rule (Appendix C.1) the database of 17 October 2026 holds 278 trials, with a benchmark of 0.60 on zero-reference moments (0.52 above cash, the basis the platform has scored on since 18 October 2026); across the 173 variants the two sets of scores differ by at most 0.061, and three variants that reach 0.90 with $K = 219$ fall below it with the platform's count (to between 0.88 and 0.89). Treating catalog entries as independent trials overstates the effective number of trials, since variants of one strategy are correlated; raw K is a modelling choice, not a validated search correction, and Appendix C.3 treats the score as the platform's model output rather than as a corrected probability. The values carry six decimals: among the 173 variants there are 156 distinct values, 9 equal to one exactly, and 89 that round to 1.00 at two decimals.

C.3 The Deflated Sharpe Ratio and the SSR together

The Deflated Sharpe Ratio and the SSR answer different questions, and Table 3 shows that they classify the catalog differently at the thresholds the platform uses. Of the 173 variants, 146 have a Deflated Sharpe Ratio (219 trials) of at least 0.90, the platform's cutoff below which it flags a strategy as fragile. Of those 146, 53 have an SSR of at least 0.5 and 93 fall below that descriptive threshold. Two variants have a Deflated Sharpe Ratio below 0.90 and an SSR of at least 0.5, and 25 fall below both thresholds. With the platform's current count of trials (Appendix C.1) the four cells would hold 53, 90, 2 and 28. The chosen cutoffs produce different classifications in this selected catalog; neither classification establishes prospective credibility. The Credibility-Adjusted SSR of the source paper is numerically close to the SSR and ranks the catalog almost identically (rank correlation above 0.99999), because the Probabilistic Sharpe Ratio at zero is at least 0.998 for every variant with a history this long.

C.4 Resolution at the displayed precision

At two decimals the SSR takes 52 distinct values across the 173 variants and the Deflated Sharpe Ratio 28, with 89 variants rounding to 1.00. The Deflated Sharpe Ratio retains more information than its rounded display suggests (156 distinct values at six decimals, of which 9 are exactly one), so the comparison is one of resolution at the precision commonly displayed rather than of the underlying statistics. Above cash the Deflated Sharpe Ratio saturates less: 29 variants round to 1.00, and at two decimals it takes 44 distinct values against the SSR's 48.

Appendix D. Full results

D.1 Distribution and resolution

Table D.1 summarises the distribution of the full-history SSR across the 173 variants. Values run from 0.09 to 0.94 with a median of 0.45; 55 variants have an SSR of at least 0.5, 6 at least 0.7, and 13 fall below 0.3. Figure D.1 shows the distribution against the two main benchmarks: the S&P 500 sits at 0.32 and the 60/40 blend at 0.34, both in the bottom quartile of the catalog.

Table D.1. Distribution of the full-history SSR, 173 published variants, completed months through August 2026.

Statistic	Value
Minimum	0.09
10th percentile	0.31
25th percentile	0.37
Median	0.45
75th percentile	0.52
90th percentile	0.61
Maximum	0.94
Distinct values at two decimals	52 (Deflated Sharpe Ratio: 28)
Variants at or above 0.5 / 0.7	55 / 6

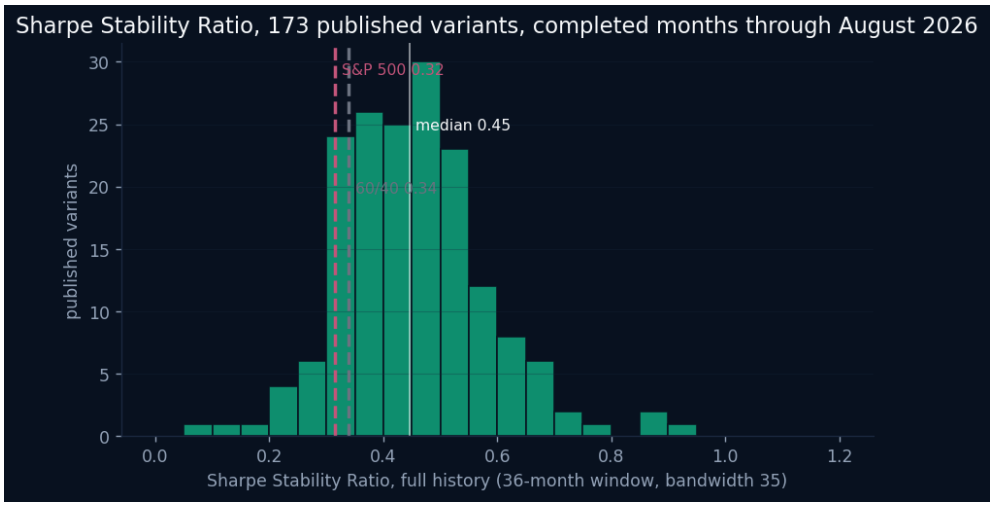


Figure D.1. Distribution of the full-history SSR across the 173 published variants (histories start between 1900 and 2008 and end in August 2026; 36-month window, bandwidth 35), with the S&P 500 (VFINX, from 1976) and the 60/40 blend (from 1987) marked.

D.2 Relation to level and to credibility

Table 2 reports Spearman rank correlations between SSR and the companion statistics. The correlation with the Sharpe ratio is 0.64, indicating imperfect rank concordance. Figure 2 shows the joint distribution with a few named strategies. The correlations with the Deflated Sharpe Ratio and the Probabilistic Sharpe Ratio are 0.59 and 0.46; their differences from the Sharpe correlation may reflect sample length, higher moments and numerical ties. SSR is only weakly related to the level of return (0.15 with the compound annual growth rate) and moderately to the depth of the worst drawdown (0.54; shallower drawdowns go with higher SSR).

D.3 Groups

Table D.2 gives medians by strategy type, rebalancing frequency, leverage and risk category, with every category shown so that the counts reconcile. Tactical variants have a higher median SSR than static ones (0.46 against 0.39), and the difference holds at both quartiles. Among frequency groups with more than one variant, monthly-rebalanced strategies have the highest median (0.47); daily strategies and annually rebalanced allocations (both 0.36) the lowest; the single weekly variant scores 0.54. Leveraged variants have a slightly lower median than unleveraged ones (0.42 against 0.46). By the platform's risk category the medians are 0.49, 0.46 and 0.39 for conservative, moderate and aggressive variants. These ratios alone do not identify the contribution of numerator and denominator; for the record, the median annualised mean rolling Sharpe ratio is 1.20, 1.08 and 0.92 in the same three groups, so the aggressive group has both a lower average rolling Sharpe ratio and a lower SSR.

Table D.2. Median full-history SSR by group (interquartile range in brackets), 173 published variants.

Group	n	Median SSR
Tactical	142	0.46 [0.38, 0.54]
Static	31	0.39 [0.33, 0.47]
Monthly rebalancing	121	0.47 [0.39, 0.54]
Weekly rebalancing	1	0.54
Quarterly rebalancing	10	0.42 [0.35, 0.46]
Daily rebalancing	24	0.36 [0.32, 0.42]
Annual rebalancing	17	0.36 [0.33, 0.49]
Unleveraged	121	0.46 [0.37, 0.53]
Leveraged	52	0.42 [0.36, 0.50]
Conservative risk	16	0.49 [0.43, 0.52]
Moderate risk	105	0.46 [0.38, 0.54]
Aggressive risk	46	0.39 [0.33, 0.44]
No risk category	6	0.50 [0.43, 0.54]

D.4 The extremes and widely followed strategies

Table D.3 lists the twelve highest and twelve lowest full-history SSR values. The top is dominated by monthly momentum rotations with diversified defensive legs: the Momentum-Correlation Triplet family (0.86 to 0.94), Pragmatic Asset Allocation (0.73), the DMS Triad family (0.66 to 0.72), two Vitral multi-asset momentum variants (0.66 and 0.67) and the KISS momentum blends (0.58 to 0.65), together with a daily leveraged rotation from 2002 (Low-Initiative LETF v2, 0.77) and a buffered TQQQ quadrant rule (0.67); the KISS range includes members below the top-twelve cutoff of 0.65. The bottom is a mix of single-asset trend rules on bonds, gold and real estate (0.11 to 0.30), daily volatility-regime strategies (0.20 to 0.22), a gold-based cross-asset rule (0.09) and static allocations (an all-equity US allocation at 0.27, a static leveraged three-asset mix at 0.29 and a global 60/40 at 0.30). Of the four century-long series, Paired Switching and UIS from 1920 and the all-equity US allocation, which now starts in 1900, sit in the bottom twelve on full history (165th, 167th and 166th); the classic 60/40 from 1923 ranks 158th of 173 with an SSR of 0.31.

The catalog minimum is an estimator artefact rather than instability in the usual sense. From December 1996 to November 1999 the gold-based cross-asset rule held a near-riskless position whose monthly returns barely varied (standard deviation 0.03 percent in the window ending November 1999), so its rolling Sharpe ratio peaks at 40.7 annualised and stays above 5 for ten months. Those few values raise the mean of the rolling series (1.44 annualised) much less than its long-run standard deviation (4.66 per month), and the SSR falls to 0.09. A rolling Sharpe ratio is not informative when the window holds a near-constant return; we report the value as computed and do not trim it.

Table D.3. Highest and lowest full-history SSR among the 173 published variants. Sharpe is the annualised monthly-mean ratio with a zero cash reference; DSR the Deflated Sharpe Ratio of Appendix C.2; the interval is the 95 percent percentile moving block bootstrap interval with 36-month blocks.

Variant	From	Sharpe	DSR	SSR	95% interval	Turnover
Triplet, standard	1987	1.21	1.000	0.94	[0.43, 1.02]	4.0
Triplet, plus	1985	1.25	1.000	0.86	[0.41, 1.05]	4.8
Triplet, 1.5x leverage	1987	1.07	0.998	0.86	[0.38, 0.95]	4.9
Low-Initiative LETF v2	2002	0.82	0.860	0.77	[0.30, 0.96]	15.4
Pragmatic Asset Allocation	1985	1.18	1.000	0.73	[0.39, 0.94]	2.6
DMS Triad, plus	1985	1.17	1.000	0.72	[0.38, 0.86]	3.1
DMS Triad	1985	1.24	1.000	0.70	[0.40, 0.89]	3.1
Vitral MAM, ensemble	1985	1.14	0.999	0.67	[0.35, 0.82]	2.7
TQQQ Quadrant, buffered	1996	0.96	0.979	0.67	[0.28, 1.05]	0.9
Vitral MAM, low protection	1985	1.29	1.000	0.66	[0.37, 0.96]	3.9
DMS Triad, plus plus	1985	1.09	0.999	0.66	[0.35, 0.82]	3.1
KISS Momentum 70/30	1985	1.24	0.999	0.65	[0.38, 0.99]	2.6
…						
Global 60/40	1985	0.76	0.851	0.30	[0.13, 0.50]	0.0
SMA Trend, REITs	1990	0.63	0.576	0.30	[0.15, 0.52]	1.8
UPRO/ZROZ/GLD static	1985	0.76	0.856	0.29	[0.14, 0.58]	0.2
Paired Switching SPY/TLT	1920	0.72	0.908	0.28	[0.18, 0.37]	2.9
US stock market, 100%	1900	0.63	0.694	0.27	[0.18, 0.34]	0.0
UIS	1920	0.72	0.897	0.24	[0.18, 0.36]	2.1
Regime Detector	1990	0.67	0.683	0.22	[0.08, 0.49]	2.2
VIX Shield, leveraged	1990	0.64	0.634	0.20	[0.06, 0.64]	4.7
Regime Detector, smart leverage	1990	0.62	0.580	0.20	[0.06, 0.48]	2.2
SMA Trend, long Treasuries	1990	0.56	0.454	0.18	[0.08, 0.45]	1.9
SMA Trend, gold	1990	0.47	0.250	0.11	[−0.01, 0.34]	2.1
Gold Cross-Asset	1985	0.69	0.728	0.09	[0.07, 0.43]	1.6

Table D.4 shows the same statistics for a set of widely followed strategies, with the components of the ratio for two of them. Hybrid Asset Allocation, the highest-Sharpe strategy in the table at 1.50, has a full-history SSR of 0.48: all 595 of its rolling estimates are positive, with a minimum annualised value of 0.26, and its annualised mean rolling Sharpe ratio is 1.58, but the HAC standard deviation of the rolling series (0.95 in per-period units) is five times its naive standard deviation (0.18), and it is the HAC figure that the ratio divides by. Its SSR on the window since 2000 is 0.87. The Momentum-Correlation Triplet has a lower mean rolling Sharpe ratio (1.18 annualised) and a much lower HAC standard deviation (0.37), which is what puts it at the top. Global Equities Momentum, with a Sharpe ratio of 1.10, has an SSR of 0.42 and a bootstrap proportion of 0.82 at or below 0.5. The classic 60/40 has a Sharpe ratio comparable to several tactical strategies and a lower SSR, which is the pattern the group medians summarise;

the Golden Butterfly, at 0.49, is an exception, above the tactical median of 0.46, as is All Weather, another static allocation, at 0.51.

Table D.4. Selected widely followed strategies. Sharpe and DSR are full-history; SSR is shown for the full history and for windows starting in January 2000 and January 2008; the final column is the proportion of 36-month-block bootstrap replicates at or below 0.5 for the full-history estimate, an uncentered proportion whose one-sided reading depends on the method's coverage (Appendix A.3).

Strategy	From	Sharpe	DSR	SSR full	2000 on	2008 on	$q \le 0.5$
HAA, Hybrid AA	1974	1.50	1.000	0.48	0.87	0.91	0.22
BAA-G12, Bold AA	1985	1.31	1.000	0.61	0.51	0.50	0.20
DAA-G12, Defensive AA	1985	1.31	1.000	0.52	0.48	0.48	0.48
PAA, Protective AA	1985	1.35	1.000	0.50	0.43	0.43	0.43
LAA, Lethargic AA	1985	1.21	1.000	0.62	0.69	0.77	0.45
Vitral Multi-Asset Momentum	1985	1.31	1.000	0.61	0.57	0.53	0.30
Momentum-Corr. Triplet	1987	1.21	1.000	0.94	0.90	0.77	0.12
GTAA-5, Global Tactical AA	1985	1.18	1.000	0.48	0.41	0.85	0.60
Accelerating Dual Momentum	1985	1.07	0.999	0.43	0.50	0.60	0.76
GEM, Global Eq. Momentum	1985	1.10	0.999	0.42	0.38	0.52	0.82
S&P 500 200-day trend	1990	0.97	0.983	0.46	0.60	0.57	0.75
Century Momentum	1928	0.95	0.999	0.41	0.64	0.67	0.95
Golden Butterfly	1985	1.16	0.999	0.49	0.53	0.62	0.61
All Weather	1972	1.07	1.000	0.51	0.42	0.38	0.66
Classic 60/40	1923	0.79	0.979	0.31	0.37	0.62	1.00
Buy the Dip (TQQQ sleeve)	1985	1.05	0.987	0.41	0.91	1.05	0.55

D.5 Benchmarks and the evaluation window

Table D.5 reports the SSR of the benchmarks on the full available history and on windows starting in January 2000 and January 2008, both ending in August 2026. The S&P 500 scores 0.32 on the full history and 0.29 on the 2000 window, and 153 of the 166 published variants with data over that window exceed it. On the 2008 window the S&P 500 scores 0.74, the Nasdaq 100 0.88 and the 60/40 blend 0.60, and 57 of 172 variants exceed the S&P 500. Figure 3 illustrates the two rolling Sharpe paths behind one pair of numbers, Hybrid Asset Allocation and the S&P 500: the first never crossed zero and the second spent 2002 to 2004 and late 2008 to 2010 below it, before holding high for a decade. The figure shows the paths; it does not measure the return cost of defensive positioning, and Table D.6 does not either.

Table D.5. SSR of benchmark series on three windows, all ending in August 2026. The 2000 and 2008 windows are shown only where the series covers them from the start.

Series	From	Sharpe	SSR full	2000 on	2008 on
S&P 500 (VFINX)	1976	0.82	0.32	0.29	0.74
60/40 (VFINX/VBMFX)	1987	0.96	0.34	0.35	0.60
US bonds (VBMFX)	1987	1.16	0.33	0.30	0.23
Long Treasuries (VUSTX)	1986	0.56	0.24	0.19	0.12
Nasdaq 100 (QQQ)	1999	0.56	0.27	0.30	0.88
Gold (GLD)	2004	0.67	0.20	n/a	0.17
Long Treasuries (TLT)	2002	0.32	0.15	n/a	0.11
Variants above the S&P 500				153 of 166	57 of 172
Variants above 60/40				132 of 166	98 of 172

Figure D.2 plots each variant's SSR on the 2000 window against its SSR on the 2008 window for the 166 variants that have both. The rank correlation between the two windows is 0.53, and between the full history and the 2008 window 0.28 (0.76 for the 2000 window, which shares most of its history with the full sample). Relative rankings change between overlapping evaluation windows, so group comparisons need common calendar dates; Table D.6 uses common subperiods, although its cohorts still differ in size and, within the first period, in history length.

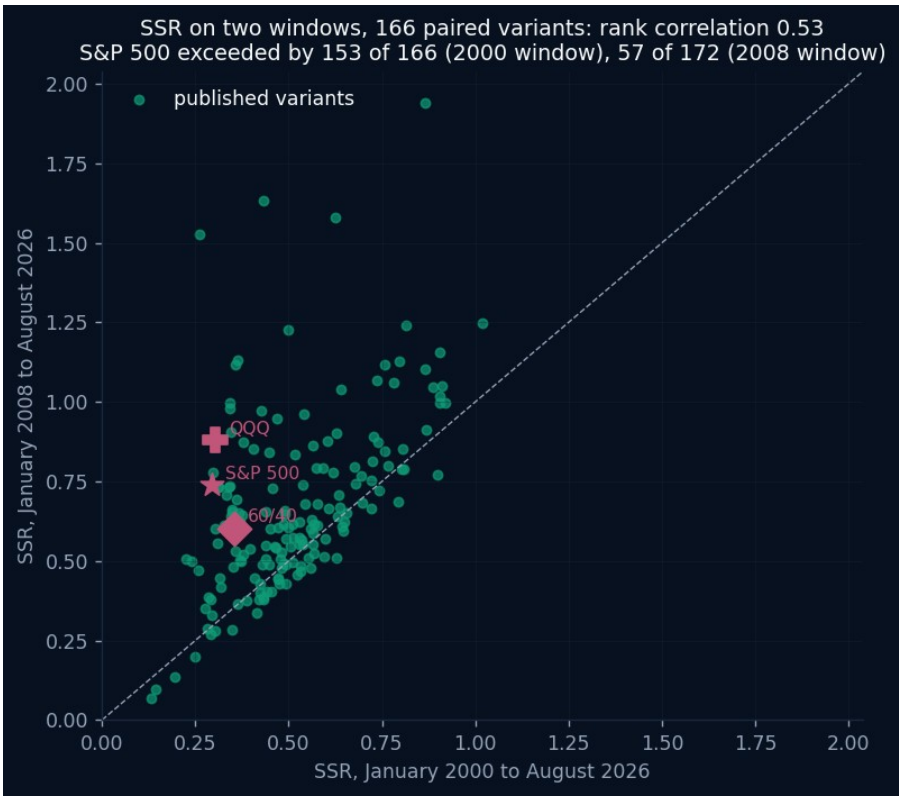


Figure D.2. SSR on the January 2000 to August 2026 window against SSR on the January 2008 to August 2026 window, 166 paired variants, benchmarks marked. The counts in the title refer to the 166-variant 2000 cohort and the 172-variant 2008 cohort of Table D.5.

D.6 Subperiods

To examine whether the cross-sectional ordering persists, we split the calendar into four subperiods, before 2000, 2000 to 2009, 2010 to 2019, and January 2020 to August 2026, and recompute the SSR inside each for every published variant with at least 72 monthly returns in it. The cohorts hold 161, 172, 173 and 174 variants; the last includes the 83-month variant excluded from the full-history tables. The last subperiod holds 80 completed monthly returns and 45 rolling estimates, the 2010s exactly 120 returns for every variant, and the pre-2000 period between 76 and 1,199 returns depending on the variant. The subperiods are therefore not four equivalent decades. Table D.6 reports the medians, the benchmarks and the cross-sectional rank correlations between subperiods with their pairwise sample sizes.

Table D.6. SSR by subperiod: medians across published variants (72-month minimum, so levels are not comparable with the full-history values), medians by strategy type, benchmark values (TLT's 2000s value rests on 89 returns), and Spearman rank correlations of the cross-section between subperiods (pairwise n in brackets).

	Before 2000	2000 to 2009	2010 to 2019	2020 to Aug 2026
Variants with an estimate	161	172	173	174
Median SSR, all	0.63	0.78	0.98	0.82

	Before 2000	2000 to 2009	2010 to 2019	2020 to Aug 2026
Median SSR, tactical	0.63	0.89	0.88	1.01
Median SSR, static	0.68	0.35	1.58	0.61
S&P 500 (VFINX)	0.52	0.15	1.63	0.81
60/40 (VFINX/VBMFX)	0.74	0.23	1.60	0.59
US bonds (VBMFX)	0.96	1.05	0.78	−0.03
Long Treasuries (TLT, from Aug 2002)	n/a	3.38	0.80	−0.47
Nasdaq 100 (QQQ)	n/a	0.13	2.23	0.70
Rank correlation with before 2000		0.08 (161)	−0.12 (161)	−0.22 (161)
Rank correlation with 2000 to 2009			−0.38 (172)	−0.09 (172)
Rank correlation with 2010 to 2019				0.11 (173)

Between adjacent subperiods the rank correlation of the cross-section is 0.08, −0.38 and 0.11. Higher estimated SSR in the 2000s is associated with lower estimated SSR in the 2010s; this is an imperfect association rather than a complete reversal, and it is not an artefact of the bandwidth: the 2000s-to-2010s correlation is −0.37, −0.37, −0.38 and −0.28 under $L=5$, 15, 35 and 71 (the bandwidth is capped at one less than the number of rolling estimates, which binds for two variants in the 2000s under the 71-lag specification). The benchmarks move with it. The S&P 500 goes from 0.15 to 1.63 between the two decades and the 60/40 blend from 0.23 to 1.60, while the median static variant goes from 0.35 to 1.58 and the median tactical variant hardly moves (0.89 and 0.88); in the 2020 window the ordering of the two types reverses again (1.01 against 0.61). The results are consistent with period dependence in estimated SSR, but differences in samples, estimators and estimation uncertainty prevent a stronger comparison with the hedge fund findings of Bajo Traver and Rodriguez Dominguez (2026), whose average subperiod rank correlation is 0.31.

Several explanations remain to be separated, and the design here does not separate them: finite-sample variability of the estimated denominator when the period holds few rolling estimates (at most 45 in the last subperiod against a 35-lag bandwidth), the removal of between-regime variation when the period is short, the restarting of rolling windows at each boundary, unequal history lengths across variants within a period, and genuine changes in how strategies behaved. Short subperiods create uncertainty in both levels and rankings. Ordinary standard errors for these rank correlations would also be inappropriate, because the variants share returns and belong to related families; we report the correlations as descriptive statistics with their sample sizes.

D.7 Level and order: the permutation benchmark in full

Appendix E.2 left open how much of a variant's SSR would arise from rolling estimation alone. A direct benchmark keeps each series' monthly returns and discards their order. Reordering a series' own returns at random leaves its mean, its standard deviation and its full-sample Sharpe ratio unchanged, and removes serial dependence, volatility clustering and regime structure. For every series we draw $B=10,000$ random orderings, recompute the SSR on each with the

estimator of Appendix A.1, and report the mean of the reordered values, the permutation multiple (the real SSR divided by that mean) and the one-sided permutation p-value $(b+1)/(B+1)$, where b counts the orderings whose SSR is at least the real one (Phipson and Smyth, 2010). The comparison was suggested by Randy Harris of Dual Momentum Systems.

The reordered level is close to proportional to the Sharpe ratio and depends on the length of the history. Under independent normal monthly returns over 499 months, the median history in the sample, the mean SSR is 0.20, 0.39, 0.58 and 0.75 at annualised Sharpe ratios of 0.5, 1.0, 1.5 and 2.0, between 0.41 and 0.37 times the Sharpe ratio. Across the 173 variants a regression of the reordered mean on the Sharpe ratio through the origin has a slope of 0.39 (Pearson correlation 0.97). At a Sharpe ratio of 1.0 the mean is 0.80 at the 120-month reporting gate (standard deviation 0.64), 0.50 at 220 months, the shortest history in the sample, 0.39 at 499 months and 0.36 at 1,519 months, the longest, consistent with the downward small-sample bias of kernel long-run variance estimates when the number of rolling estimates is not large relative to the window. SSR levels computed over histories of very different length are therefore not directly comparable, even with the same window and bandwidth. In this sample the effect on rankings is moderate, because most histories run between 439 and 630 months (tenth and ninetieth percentiles): the rank correlation between history length and the ratio of the reordered mean to the Sharpe ratio is −0.52, and between history length and the SSR itself 0.07. The four century-long series of Appendix D.4 owe their low placing mainly to their Sharpe ratios; their multiples are 1.01 (Paired Switching), 0.90 (UIS), 1.05 (classic 60/40) and 1.14 (the all-equity US allocation from 1900).

Across the catalog the permutation multiple runs from 0.31 to 2.12, with a median of 1.12 and an interquartile range of 0.99 to 1.22; 13 variants have a multiple of at least 1.5, 2 of at least 2.0, and 46 at or below 1.0. A multiple above one means that the rolling Sharpe ratio was steadier, relative to its mean, than the same returns in random order would make it; a multiple below one means that good and bad months were more clustered in time than chance ordering produces. Twenty-five variants have a permutation p-value below 0.05 and five below 0.01, against about 8.7 expected at 5 percent if no series had temporal structure and the tests were independent. The tests are not independent, since variants of one family share returns, and no variant remains below the threshold after a Benjamini and Hochberg (1995) false discovery rate adjustment at 5 percent. The excess of small p-values across the catalog is suggestive; no individual variant's multiple is established by this test.

The multiple removes most of the Sharpe level from the ranking. Its rank correlation with the Sharpe ratio is −0.01, against 0.64 for the SSR, and its rank correlation with the SSR is 0.70. Its rank correlation with the ratio of the SSR to the Sharpe ratio is 0.97: because the reordered mean is nearly proportional to the Sharpe ratio, the multiple is in practice the SSR-to-Sharpe ratio, rescaled and adjusted for history length and for the shape of the return distribution, and among histories of similar length it can be approximated from those two statistics.

Table D.7 reports the benchmark for a subset of the strategies of Tables D.3 and D.4 and the two main benchmarks. The Momentum-Correlation Triplet, the DMS Triad and Pragmatic Asset Allocation, near the top of the SSR ranking, also have multiples well above one (2.00, 1.49 and 1.63), so their SSRs are not explained by their Sharpe levels alone; the twelve variants with the highest SSR in Table D.3 all have multiples between 1.32 and 2.12, and eleven of them have permutation p-values below 0.05. Hybrid Asset Allocation, the highest-Sharpe strategy in Table

D.4, has a multiple of 0.84 on its full history: its SSR of 0.48 is below the mean of 0.57 for its own returns in random order, consistent with the long swings in its rolling Sharpe ratio shown in Figure 3. The S&P 500, the 60/40 blend and the three static allocations in the table sit between 0.94 and 1.27; the highest, All Weather, has a p-value of 0.079, and the other four have p-values of 0.19 or more.

Table D.7. Permutation benchmark for selected series: the full-history SSR, the mean SSR of 10,000 random orderings of the same monthly returns (Random), their ratio (Multiple) and the one-sided permutation p-value, with the multiple and p-value on the window from January 2000, which every series in the table covers from its start.

Series	Sharpe	SSR	Random	Multiple	p	2000 on	p (2000)
Momentum-Corr. Triplet	1.21	0.94	0.47	2.00	0.002	1.90	0.011
DMS Triad	1.24	0.70	0.47	1.49	0.027	1.44	0.081
Pragmatic AA	1.18	0.73	0.45	1.63	0.012	1.42	0.087
LAA, Lethargic AA	1.21	0.62	0.46	1.35	0.069	1.62	0.036
BAA-G12, Bold AA	1.31	0.61	0.52	1.17	0.190	1.00	0.409
DAA-G12, Defensive AA	1.31	0.52	0.51	1.02	0.398	1.04	0.364
HAA, Hybrid AA	1.50	0.48	0.57	0.84	0.796	1.70	0.031
GTAA-5	1.18	0.48	0.44	1.10	0.272	0.96	0.468
GEM	1.10	0.42	0.42	1.00	0.447	1.09	0.308
Golden Butterfly	1.16	0.49	0.42	1.16	0.194	1.29	0.136
All Weather	1.07	0.51	0.40	1.27	0.079	1.10	0.299
Classic 60/40	0.79	0.31	0.29	1.05	0.296	1.14	0.256
Paired Switching	0.72	0.28	0.27	1.01	0.435	1.10	0.297
UIS	0.72	0.24	0.27	0.90	0.802	1.08	0.316
S&P 500 (VFINX)	0.82	0.32	0.31	1.02	0.407	1.13	0.267
60/40 (VFINX/VBMFX)	0.96	0.34	0.36	0.94	0.563	1.13	0.260

The multiple depends on the evaluation period as much as the SSR does. On the window from January 2000 the median multiple across the 166 variants with data is 1.24 and 32 have a p-value below 0.05; the rank correlation with the full-history multiple is 0.59. Hybrid Asset Allocation moves from 0.84 to 1.70 ($p = 0.031$), approaching the Triplet's 1.90, while the DMS Triad stays close to its full-history value (1.49 and 1.44). Whether a strategy's rolling Sharpe ratio looks more or less orderly than chance depends on which decades the window holds. The benchmark describes the realised series: it conditions on their returns and does not test whether those returns, or the rules that produced them, would recur.

D.8 Turnover

The unadjusted association between reported annual turnover and SSR is positive: the rank correlation is 0.26 across the 173 variants. Table D.8 splits the variants into turnover terciles with a deterministic tie policy (turnover, then variant identifier). The lowest tercile (turnover

up to 1.3 times the portfolio per year) has a median SSR of 0.42, the middle tercile (1.3 to 3.0) 0.42 and the highest tercile (3.0 to 15.4) 0.48, with median full-sample Sharpe ratios of 0.99, 1.00 and 1.05. Figure D.3 shows the relation. The terciles differ substantially in composition: all 31 static variants and all 27 annually or quarterly rebalanced variants are in the lowest tercile, while the top tercile is entirely tactical and holds 23 of the 52 leveraged variants. Within tactical variants the rank correlation between turnover and SSR is 0.19 (142 variants), and within monthly tactical variants 0.14 (117 variants), so static membership does not explain the whole association, but these descriptive comparisons do not isolate a turnover effect: strategy type, rebalancing frequency, leverage and asset exposures all move with turnover. Persistent asset exposures may contribute to period differences in rolling Sharpe ratios, and a defensive leg may reduce them; testing that explanation requires holdings and return attribution across market conditions, which this paper does not attempt.

Table D.8. Turnover terciles, 173 published variants, full history, with composition.

Tercile	Annual turnover range	n	Median SSR	Median Sharpe	Tactical / static	Monthly / daily / other
Lowest	0.0 to 1.3	57	0.42	0.99	26 / 31	23 / 7 / 27
Middle	1.3 to 3.0	57	0.42	1.00	57 / 0	49 / 8 / 0
Highest	3.0 to 15.4	59	0.48	1.05	59 / 0	49 / 9 / 1

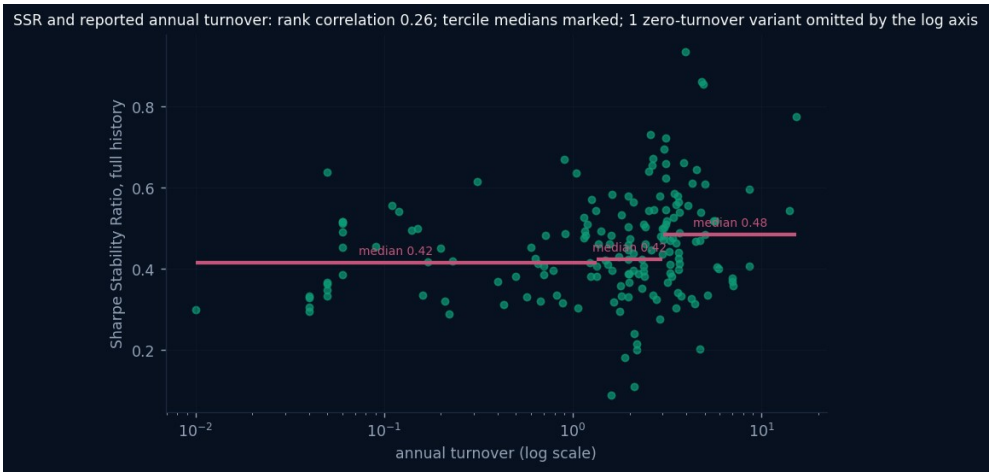


Figure D.3. Full-history SSR against reported annual turnover (log scale), tercile medians marked; the one variant with zero reported turnover is omitted by the log axis. Rank correlation 0.26.

Appendix E. Robustness and uncertainty

E.1 Window, bandwidth and sample sensitivity

Table E.1 reports the sensitivity of the cross-section to the rolling window and to the HAC bandwidth. The first four rows change the window and the bandwidth together ($L = \tau - 1$), as the definition does; the next two hold the bandwidth at 35 and change the window alone. Rankings are similar across windows of 24, 36, 48 and 60 months (rank correlation 0.92, 0.98 and 0.98 between adjacent lengths; 0.82 between 24 and 60) and levels rise with the window

(medians 0.42 to 0.48), as in the source paper. The bandwidth moves levels much more: the median SSR is 0.85 with 5 lags, 0.56 with 15, 0.45 with 35 and 0.42 with 71, and the sample-size rule gives 0.85. Rank correlations with the 35-lag baseline are 0.96, 0.99, 0.93 and 0.96, but individual variants can move substantially: one moves 55 places between the 35- and 71-lag specifications. Comparisons should state both window and bandwidth choices, and high aggregate rank correlations in this sample do not guarantee agreement for individual strategies or other datasets.

Table E.1. Sensitivity of the cross-section to the rolling window and the HAC bandwidth, 173 published variants.

Specification	Median SSR	Rank correlation with baseline
Window 24 months, bandwidth 23	0.42	0.92 (with 36/35)
Window 36 months, bandwidth 35 (baseline)	0.45	1.00
Window 48 months, bandwidth 47	0.45	0.98 (with 36/35)
Window 60 months, bandwidth 59	0.48	0.93 (with 36/35)
Window 24 months, bandwidth 35	0.40	0.94
Window 60 months, bandwidth 35	0.54	0.97
Window 36 months, bandwidth 5	0.85	0.96
Window 36 months, bandwidth 15	0.56	0.99
Window 36 months, bandwidth 71	0.42	0.93
Window 36 months, sample-size rule (4 to 7 lags)	0.85	0.96

Two sample choices matter little for the aggregates: the exclusion of the revised-data variants and the treatment of the partial month. Including the three excluded variants of the revised-data strategy gives 176 variants, a median of 0.45, 58 variants at or above 0.5, a rank correlation with the Sharpe ratio of 0.65 and a 2000s-to-2010s rank correlation of −0.38. Keeping the observations after August 2026 (September and the partial October), as the platform's strategy pages do, leaves the median at 0.45 and the count at or above 0.5 at 55, and gives a rank correlation of 0.9998 with the values through August.

E.2 Bootstrap intervals

Figure E.1 shows the 95 percent percentile intervals for the twelve highest and twelve lowest variants. All 173 percentile intervals exclude zero with 12-month blocks, and 172 with 36-month blocks; the exception is the trend rule on gold, whose interval reaches −0.01. This concerns positive mean rolling performance: it does not test whether temporal stability exceeds what rolling estimation on a constant positive Sharpe ratio would produce. Appendix D.7 addresses that question with a permutation benchmark. No variant's 5th bootstrap percentile exceeds 0.5, and 10 variants have percentile intervals that lie entirely below 0.5. The median interval width is 0.44.

The bootstrap distributions are not centred on the point estimates for the steadiest strategies. Across the catalog the median ratio of bootstrap mean to point estimate is 1.01, but for the Momentum-Correlation Triplet the bootstrap mean is 0.67 against a point estimate of 0.94, and

its percentile interval [0.43, 1.02] and basic interval [0.85, 1.44] give opposite answers about 0.5; for Hybrid Asset Allocation the bootstrap mean is 0.58 against 0.48, with intervals [0.42, 0.83] and [0.13, 0.55]. Block resampling alters the long-run structure of a series, which may explain the downward shift for the steadiest strategies; the intervals are reported as descriptive and no coverage is claimed. Marginal intervals also say nothing about the uncertainty of differences between neighbours or of ranks, which would require synchronised resampling across strategies. The qualitative pattern, intervals above zero for all but one variant and no 5th percentile above 0.5, resembles the source paper's hedge fund findings, although the estimators, bootstrap designs and significance levels differ.



Figure E.1. SSR point estimates with 95 percent percentile moving block bootstrap intervals (36-month blocks, 2,000 replicates), twelve highest and twelve lowest published variants.

E.3 Results above cash

Section 5.6 summarises the results with every monthly return measured above US dollar cash, equation (10). This section gives the detail. The cash index is the platform's daily US dollar cash level (a Treasury bill fund chain from 1900), read at the last observation on or before each month-end and at most seven days old; each strategy and benchmark return is reduced by the cash return between the same two month-ends, and the 60/40 blend is formed from the two funds' above-cash returns. The estimator, sample, seeds and gates are those of Appendix A. The Deflated Sharpe Ratio uses the platform's stored above-cash moments (monthly Sharpe ratio, skewness and kurtosis of $r_t - c_t$ over the full stored history) with $K = 219$, which gives a no-skill benchmark of 0.46 annualised; the live universe of Appendix C.1 gives 0.52 on the same moments.

Distribution. Above cash the SSR runs from 0.01 to 0.78, with quartiles of 0.26 and 0.38, a median of 0.32 and a 90th percentile of 0.49; 17 variants reach 0.5, one reaches 0.7, 51 reach 0.37

and 70 fall below 0.3. At two decimals the SSR takes 48 distinct values and the Deflated Sharpe Ratio 44, with 29 variants rounding to 1.00. The split that would flag exactly as many variants as 0.5 does with a zero reference (118) lies between 0.362 and 0.363, so the platform's 0.37, calibrated on its own published variants, carries over to this sample within four variants.

Joint classification. Of the 112 variants with a Deflated Sharpe Ratio of at least 0.90, 46 have an SSR of at least 0.37 and 66 fall below it; of the other 61, 5 reach 0.37 and 56 do not. At the 0.5 cutoff the four cells hold 16, 96, 1 and 60. With the live universe of Appendix C.1 (278 trials, above-cash moments) the cells at 0.37 hold 42, 47, 9 and 75.

Groups and turnover. The median above-cash SSR is 0.33 for tactical and 0.28 for static variants (interquartile ranges [0.27, 0.39] and [0.25, 0.33]); by rebalancing frequency, 0.33 monthly, 0.30 quarterly, 0.29 daily and 0.27 annual. The turnover terciles of Table D.8 have medians of 0.30, 0.31 and 0.38, and all 31 static variants remain in the lowest. Unlike the zero reference, leveraged variants have a slightly higher median than unleveraged ones (0.33 against 0.32), and the conservative risk category the lowest (0.27).

Widely followed strategies. Table E.3 repeats Table D.4 above cash, with the permutation multiple and p-value. The Bold Asset Allocation rotation, the Lethargic Asset Allocation and the GTAA-5 fall well below their zero-reference values, the Defensive and Protective rotations less; HAA rises; the Golden Butterfly sits just above the tactical median (0.34 against 0.33) and All Weather well below it (0.23).

Windows, bandwidths and subperiods. The median above-cash SSR is 0.30, 0.32, 0.34 and 0.35 for windows of 24, 36, 48 and 60 months, and 0.62, 0.41, 0.32 and 0.31 for bandwidths of 5, 15, 35 and 71 lags; rank correlations with the baseline run from 0.92 to 0.99, and one variant moves 57 places between the 35- and 71-lag specifications. Table E.4 gives the subperiods. The 2000s-to-2010s rank correlation is −0.37 (−0.38 with the zero reference); the correlation between the first two subperiods becomes negative (−0.18) and between the last two rises to 0.26.

Uncertainty and the permutation benchmark. With 36-month blocks 170 of the 173 percentile intervals exclude zero (the exceptions are the trend rules on gold and long Treasuries and the gold-based cross-asset rule), no 5th percentile exceeds 0.5, 55 intervals lie entirely below 0.5 and the median width is 0.39. The permutation multiple has a median of 1.08 and an interquartile range of 0.96 to 1.22; it exceeds one for 115 variants, 12 reach 1.5 and one reaches 2.0. Its rank correlation with the zero-reference multiple is 0.60, and 98 variants have a multiple above one under both references. On the window from January 2000 the median multiple is 1.30 and 37 variants have a p-value below 0.05.

Table E.2. Highest and lowest full-history SSR above cash among the 173 published variants, with the zero-reference SSR and its rank, Rank (0), for comparison. Sharpe is the annualised monthly Sharpe ratio of the returns above cash; DSR the above-cash Deflated Sharpe Ratio with 219 trials; the interval is the 95 percent percentile bootstrap interval (36-month blocks).

Variant	From	Sharpe	DSR	SSR	95% interval	SSR(0)	Rank (0)
Low-Initiative LETF v2	2002	0.73	0.890	0.78	[0.25, 0.92]	0.77	4

Variant	From	Sharpe	DSR	SSR	95% interval	SSR(0)	Rank (0)
Triplet, standard	1987	0.96	0.999	0.63	[0.30, 0.80]	0.94	1
RVol Shifter, cash only	2000	0.75	0.918	0.62	[0.32, 1.03]	0.60	21
Triplet, 1.5x leverage	1987	0.90	0.996	0.61	[0.30, 0.80]	0.86	3
Golden Ratio Dual Gate	2008	1.05	0.990	0.59	[0.41, 1.47]	0.62	17
Triplet, plus	1985	0.95	0.999	0.58	[0.30, 0.68]	0.86	2
Pragmatic Asset Allocation	1985	0.91	0.997	0.57	[0.29, 0.71]	0.73	5
HAA, no QQQ	1974	1.12	1.000	0.57	[0.37, 0.82]	0.48	69
HAA, standard	1974	1.10	1.000	0.56	[0.35, 0.84]	0.48	66
HAA, simple momentum	1974	0.86	0.996	0.56	[0.29, 0.75]	0.58	26
HAA, leveraged	1974	1.04	1.000	0.54	[0.35, 0.81]	0.51	51
HAA, 1.5x leverage	1974	1.03	1.000	0.53	[0.33, 0.79]	0.48	64
…							
Permanent Portfolio, static	1960	0.48	0.542	0.20	[0.09, 0.30]	0.50	58
RP3 trend-tilted, SMA10	1972	0.54	0.711	0.20	[0.08, 0.37]	0.58	23
US stock market, 100%	1900	0.44	0.390	0.19	[0.12, 0.25]	0.27	166
VIX Shield, leveraged	1990	0.57	0.735	0.18	[0.04, 0.60]	0.20	169
Regime Detector	1990	0.55	0.687	0.18	[0.05, 0.44]	0.22	168
Paired Switching SPY/TLT	1920	0.48	0.572	0.18	[0.10, 0.26]	0.28	165
Regime Detector, smart leverage	1990	0.54	0.671	0.17	[0.04, 0.44]	0.20	170
ACA dynamic bond, gold sleeve	1985	0.45	0.451	0.16	[0.02, 0.32]	0.30	160
UIS	1920	0.39	0.226	0.15	[0.08, 0.24]	0.24	167
SMA Trend, long Treasuries	1990	0.29	0.146	0.10	[−0.02, 0.40]	0.18	171
Gold Cross-Asset	1985	0.44	0.384	0.05	[−0.05, 0.26]	0.09	173
SMA Trend, gold	1990	0.26	0.108	0.01	[−0.12, 0.22]	0.11	172

Table E.3. Selected widely followed strategies above cash: Sharpe ratio, Deflated Sharpe Ratio, SSR on the full history and on the windows from January 2000 and January 2008, permutation multiple and p-value on the full history, and the zero-reference SSR of Table D.4.

Strategy	Sharpe	DSR	SSR full	2000 on	2008 on	Multiple	p	SSR(0)
HAA, Hybrid AA	1.10	1.000	0.56	0.83	0.81	1.32	0.059	0.48
BAA-G12, Bold AA	0.96	0.999	0.36	0.36	0.29	0.94	0.543	0.61
DAA-G12, Defensive AA	0.97	0.999	0.39	0.36	0.29	1.03	0.385	0.52
PAA, Protective AA	0.96	0.999	0.42	0.36	0.31	1.11	0.259	0.50
LAA, Lethargic AA	0.84	0.988	0.37	0.61	0.62	1.13	0.234	0.62
Vitral Multi-Asset Momentum	0.89	0.996	0.40	0.43	0.36	1.14	0.229	0.61
Momentum-Corr. Triplet	0.96	0.999	0.63	0.72	0.62	1.65	0.013	0.94
GTAA-5, Global Tactical AA	0.70	0.921	0.29	0.36	0.54	1.04	0.357	0.48
Accelerating Dual Momentum	0.86	0.992	0.31	0.43	0.44	0.93	0.582	0.43
GEM, Global Eq. Momentum	0.86	0.991	0.38	0.36	0.41	1.13	0.243	0.42
S&P 500 200-day trend	0.72	0.920	0.35	0.44	0.41	1.22	0.156	0.46
Century Momentum	0.76	0.996	0.32	0.49	0.56	1.15	0.110	0.41
Golden Butterfly	0.74	0.942	0.34	0.44	0.44	1.16	0.191	0.49
All Weather	0.52	0.639	0.23	0.31	0.29	1.11	0.241	0.51
Classic 60/40	0.50	0.644	0.20	0.26	0.47	1.04	0.334	0.31
Buy the Dip (TQQQ sleeve)	0.93	0.989	0.33	0.73	0.93	0.82	0.834	0.41
S&P 500 (VFINX)	0.54		0.22	0.24	0.62	1.04	0.358	0.32
60/40 (VFINX/VBMFX)	0.64		0.25	0.25	0.45	0.98	0.486	0.34

Table E.4. SSR above cash by subperiod (72-month minimum), as Table D.6.

	Before 2000	2000 to 2009	2010 to 2019	2020 to Aug 2026
Variants with an estimate	161	172	173	174
Median SSR, all	0.35	0.56	0.93	0.73
Median SSR, tactical	0.33	0.64	0.85	0.76
Median SSR, static	0.37	0.24	1.35	0.62
S&P 500 (VFINX)	0.28	0.08	1.41	0.91
60/40 (VFINX/VBMFX)	0.45	0.12	1.28	0.60

	Before 2000	2000 to 2009	2010 to 2019	2020 to Aug 2026
Rank correlation with previous subperiod		−0.18 (161)	−0.37 (172)	0.26 (173)

Appendix F. Extended literature

F.1 Sharpe ratio inference

Lo (2002) derived the asymptotic distribution of the Sharpe ratio estimator under independent and identically distributed returns, treated stationary returns and time aggregation, and formulated the serially correlated case in HAC and GMM form. Mertens (2002) gave the estimator's variance under independent but non-normal returns, and Opdyke (2007) showed that the same expression holds under stationary and ergodic returns and extended it to comparisons of Sharpe ratios. Bailey and López de Prado (2012) built the Probabilistic Sharpe Ratio on that variance: an inferential score, under its assumptions, for whether the true Sharpe ratio exceeds a benchmark given the estimate, the sample length, and the skewness and kurtosis of returns. Ledoit and Wolf (2008) proposed HAC standard errors and a studentized time-series bootstrap for comparing Sharpe ratios under dependent returns. López de Prado, Lipton and Zoonekynd (2026) provide a closed-form variance under first-order autocorrelation and non-normality,

$$\sigma^2[\widehat{\mathrm{SR}}] = \frac{1}{T}\left(\frac{1+\rho}{1-\rho} - \frac{1+\rho+\rho^2}{1-\rho^2}\hat{\gamma}_3\,\mathrm{SR} + \frac{1+\rho^2}{1-\rho^2}\frac{\hat{\gamma}_4 - 1}{4}\,\mathrm{SR}^2\right) \tag{F.1}$$

where ρ is the lag-one autocorrelation of returns, $\hat{\gamma}_3$ the skewness and $\hat{\gamma}_4$ the kurtosis. For the monthly return series studied here the correction is small: the median lag-one autocorrelation across the 219 active variants is −0.02 (tenth and ninetieth percentiles −0.08 and +0.03), and recomputing each variant's Deflated Sharpe Ratio with this variance (using $T-1$ in the denominator, as the platform does) changes scores by between −0.016 and +0.020; two variants, both published, cross the platform's 0.90 threshold, one in each direction. The correction is therefore not material to the rankings, although it can move threshold classifications at the margin.

The Deflated Sharpe Ratio (Bailey and López de Prado, 2014) replaces the benchmark of the Probabilistic Sharpe Ratio with the expected maximum Sharpe ratio among K trials, given the cross-sectional variance of Sharpe ratios across those trials. It compares the observed Sharpe ratio with a model-based expected maximum under the assumed no-skill trial distribution. The 2014 paper already discusses dependent trials and an effective number of independent trials; the 2026 paper retains the expected-maximum benchmark while rescaling the standard error of the maximum by the standard deviation of the maximum of K standard normal variables. The platform's published score follows the 2014 formulation, which we take as given here. These statistics describe the full-sample estimate; in the terminology of Bajo Traver and Rodriguez Dominguez (2026) they are point-in-time, and say nothing about how the estimate was distributed across the sample.

F.2 The Sharpe Stability Ratio

Bajo Traver and Rodriguez Dominguez (2026) propose the Sharpe Stability Ratio (SSR) for that purpose. They treat the rolling Sharpe ratio as a time series, show that overlapping windows induce autocorrelation that a naive standard deviation understates substantially (naive-to-HAC standard deviation ratios of about 0.19 in their synthetic examples and 0.26 to 0.29 in their hedge fund examples), and define stability as the mean of the rolling series divided by a heteroskedasticity and autocorrelation consistent (HAC) estimate of its long-run standard deviation. Their empirical work covers seventeen hedge fund style indices over 29 years.

For a return series $r_1, \ldots, r_T$ and a window τ, Bajo Traver and Rodriguez Dominguez (2026) define the rolling Sharpe process $Z_t = \bar{r}_{t,\tau} / \hat{\sigma}_{t,\tau}$ for $t = \tau, \ldots, T$, with the sample mean and sample standard deviation over the trailing τ observations. The structural Sharpe Stability Ratio is

$$\mathrm{SSR}_0 = \frac{\mu_Z}{\sigma_\infty(Z)} \tag{F.2}$$

the population mean of the rolling process divided by its long-run standard deviation, with feasible analogue $\widehat{SSR}_0 = \bar{Z} / \hat{\sigma}_{\mathrm{HAC}}(Z)$. Their Proposition 3.1 shows that under independent returns a rolling statistic inherits a mechanical autocorrelation $\rho_k \approx 1 - k/\tau$ for lags $k < \tau$ and zero beyond, so that the naive standard deviation of Z captures only about one fifth of its long-run counterpart in their synthetic examples and between a quarter and three tenths in their hedge fund examples. The HAC estimator is the Newey and West (1987) form with a Bartlett kernel,

$$\hat{\sigma}^2_{\mathrm{HAC}}(Z) = \hat{\gamma}_0 + 2 \sum_{k=1}^{L} \left(1 - \frac{k}{L+1}\right) \hat{\gamma}_k \tag{F.3}$$

with $\hat{\gamma}_k$ the sample autocovariances of Z and L the bandwidth. Under stationarity, mixing, moment and long-run-variance consistency conditions the studentized statistic $\sqrt{N}\,\widehat{SSR}$, with $N = T - \tau + 1$ the number of rolling windows, is asymptotically standard normal under the boundary null, and the authors recommend a moving block bootstrap for finite-sample inference because the overlap-induced dependence makes the asymptotic approximation unreliable. Their hedge fund results range from 0.27 to 0.62 with a 36-month window and an Andrews (1991) bandwidth that selected 15 lags; the sample-size formula $\lfloor 4 (N/100)^{2/9} \rfloor$ selected 5 lags on the same data, with SSR levels 34 percent lower under Andrews and a rank correlation of 0.98 between the two. They document that rankings survive changes of window and bandwidth while levels do not, and that stability is conditional on the regime: across three subperiods of their sample the cross-sectional rank correlation of SSR averages 0.31.

SSR measures the average rolling Sharpe ratio relative to its long-run standard deviation, combining performance level, dispersion and serial dependence in one number. Two hedge fund indices with nearly identical Sharpe ratios in their sample (Merger Arbitrage at 1.73 and Equity Market Neutral at 1.72) have SSRs of 0.62 and 0.40. The authors recommend a benchmark Sharpe ratio $\mathrm{SR}^\star = 0$ for empirical use because the ratio's elasticities diverge as the benchmark approaches the mean rolling Sharpe ratio, define a Credibility-Adjusted SSR as the

product of SSR and the Probabilistic Sharpe Ratio, and note that SSR is the more useful of the two when the Probabilistic Sharpe Ratio already exceeds 0.95.

Appendix G. Version history

G.1 Changes from version 0.6

Version 0.7 recomputes every number, table and figure on the platform's backtests as re-run on 17 October 2026 and adds the results above cash. The sample definition, the estimator, its settings, the random seeds and the count of trials for the Deflated Sharpe Ratio ($K = 219$) are unchanged, and so is the cutoff: completed months through August 2026. Two things changed in the data. First, the platform audited its proxy chains: indicators now warm up on the price history available before a strategy's earliest permitted start, so most backtests begin at that start (127 of the 177 published variants start earlier, typically in January 1985 rather than early 1986), and several proxy histories were corrected, among them gold before 1999. Second, the all-equity US allocation now reaches back to February 1900, so the longest history is 1,519 months rather than 1,274. The benchmark series, observed fund prices, are unchanged.

Among the headline figures, the maximum SSR moves from 0.91 to 0.94 (still the Triplet) and the median from 0.44 to 0.45; the minimum moves from 0.15 to 0.09, an estimator artefact explained in Appendix D.4; the rank correlation with the Sharpe ratio moves from 0.61 to 0.64 and with the Deflated Sharpe Ratio from 0.55 to 0.59; 146 rather than 152 variants reach a Deflated Sharpe Ratio of 0.90, of which 53 rather than 55 have an SSR of at least 0.5; the adjacent-subperiod rank correlations move from −0.06, −0.40 and 0.10 to 0.08, −0.38 and 0.11; the permutation multiple's rank correlation with the Sharpe ratio moves from 0.05 to −0.01, and 127 rather than 129 multiples exceed one; one bootstrap interval, of the trend rule on gold, now reaches below zero. The live count of trials quoted in Section 4 and Appendix C is now that of the platform's universe without single-fund holdings (278 trials in the database of 17 October 2026 under the rule in force since 18 October, benchmark 0.60 on zero-reference moments); since 18 October 2026 the platform publishes its Sharpe ratio, Deflated Sharpe Ratio and Stability value above cash.

Version 0.7 also adds Section 5.6 and Appendix E.3, which repeat the headline results with returns measured above US dollar cash (new equation (10), Table 6, Figure 4 and Tables E.2 to E.4), a key-terms entry for that definition, the cash reference as a fourth choice to state with any SSR (Section 3.4), and a note on the near-cash episode behind the catalog minimum (Appendix D.4). The limitation on the zero cash reference is rewritten in the light of Section 5.6.

G.2 Changes from version 0.5

Version 0.6 changes the structure of the paper, not its results. The data, the sample, the estimator, every table value and every number of version 0.5 are unchanged, and the Deflated Sharpe Ratio still counts $K = 219$ trials (Appendix C.1). The main text now carries one message and four supporting findings, defines every term (Section 2), explains the statistic with a worked example (Section 3, with a new Figure 1 and Table 1 computed from the same series and the same estimator), and states the formulas as numbered displayed equations, including the permutation benchmark (equations (7) to (9)). The details moved to appendices: the

estimator, bandwidth and bootstrap (former Sections 4.1 to 4.4, now Appendix A), the data and replication (former Sections 3.1, 3.2 and Appendix A, now Appendix B), the Deflated Sharpe Ratio and its count of trials (former Sections 3.3 and 7, now Appendix C, with a new explanation of why $K = 219$), the full results tables (former Sections 5, 6.1, 6.2, 6.5 and 8, now Appendix D), robustness (former Sections 6.3 and 6.4, now Appendix E) and the literature review (former Section 2, now Appendix F). Tables 2 and 3 are the former Tables 2 and 10; Tables 4 and 5 condense the former Tables 9 and 7, which appear in full as Tables D.7 and D.6. The abstract is shorter. The equations are typeset so that the bar and hat accents print correctly.

G.3 Changes from version 0.4

Section, table and figure numbers in this note are those of version 0.5.

Version 0.5 recomputes every number, table and figure on restated data. The sample definition is unchanged: the same 219 variants, the same 177 published and 173 eligible variants, completed months through August 2026, the same exclusion, estimator, parameters and random seeds, and the same count of trials for the Deflated Sharpe Ratio ($K = 219$, Section 3.3). On 28 September 2026 the platform restated every backtest for two reasons. First, the engine now holds positions between signals and lets their weights drift with prices, where it previously applied each signal's target weights to every session; turnover is measured from the drifted weights (Section 3.2). Second, several fund histories that serve as proxies in the early years lacked distributions in the price provider's records, mainly before 1990: bond, real estate and sector funds, and the S&P 500 index fund's distributions before 1980 and capital gains from 1979 to 1986. The distributions are restored, the S&P 500 proxy until early January 1987 is the index's daily total return, and the Nasdaq 100 proxy before October 1985 is a large-growth total-return index; the S&P 500, US bond, long Treasury and 60/40 benchmarks carry the same corrections, which leave their 2000 and 2008 windows unchanged. Among the headline figures, the maximum SSR moves from 0.98 to 0.91 and the median stays at 0.44; the rank correlation with the Sharpe ratio moves from 0.67 to 0.61 and with the Deflated Sharpe Ratio from 0.61 to 0.55; 152 variants rather than 146 reach a Deflated Sharpe Ratio of 0.90, of which 55 rather than 54 have an SSR of at least 0.5; the adjacent-subperiod rank correlations move from −0.04, −0.43 and 0.08 to −0.06, −0.40 and 0.10; and the S&P 500's full-history Sharpe ratio moves from 0.75 to 0.82 and its SSR from 0.30 to 0.32. Version 0.5 also adds Section 6.5 with its Table 9 (former Tables 9 and 10 are now Tables 10 and 11), the AI use statement, and a more precise description of the reference fixture in Section 4.4.